



Brain tumor stage classification using magnetic resonance imaging (MRI) scans

Graduation Project, Part-I

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Date submitted

18 Dec 2025

Abstract

This project aims to develop a deep learning-based system to classify the stages of a brain tumor using magnetic resonance imaging (MRI) scans. While most current studies focus on detecting the presence or absence of a tumor, this project provides precise classification of tumor stages, enabling doctors to design effective, personalized, and carefully planned treatment strategies while monitoring disease progression efficiently. Accurate stage classification improves patient outcomes, supports medical decision-making, and ensures timely interventions. The system will be developed using MRI images from open-source datasets, which are labeled according to tumor stages. Open-source data ensures transparency, consistency, and reproducibility, allowing other researchers to validate and extend the methodology. The dataset will be divided into training, validation, and testing subsets to rigorously evaluate model performance and generalization to new, unseen images. Evaluation will focus on accuracy, robustness, and reproducibility, avoiding overfitting and ensuring unbiased results. The outputs will be clinically useful and scientifically credible, determining the system's readiness for real-world medical applications. Automated tumor stage classification supports healthcare professionals in making informed decisions, enhancing monitoring of tumor progression, and improving treatment planning. By emphasizing open-source data, reproducibility, and reliable AI methods, this project contributes to integrating artificial intelligence in healthcare. It provides a practical approach to improving diagnostics, treatment planning, and overall patient care for individuals diagnosed with brain tumors.

1 Table of Contents

1 Introduction	9
1.1 Overview of the Project	9
1.2 Problem Statement	9
1.3 Goal and Objectives	10
1.4 Expected Outcomes	10
1.5 Summary	10
2 Literature Review	12
2.1 Introduction	12
2.2 Background Research	12
2.3 Gaps in Literature	19
2.4 Feasibility Study	19
2.5 Summary	19
3 Methodology	22
3.1 Introduction	22
3.2 Development Approach	22
3.3 Tools and Technologies	22
3.4 Project Timeline	23
3.5 Risk Management	23
3.6 Quality Assurance	24
3.7 Dataset Description	24
3.8 Algorithm Description	25
3.9 System Architecture	26
3.10 Summary	27
4 Requirement Gathering	29
4.1 Introduction	29
4.2 Stakeholder Identification	29
4.3 Methods of Requirements Gathering	30
4.4 System Requirements	43
4.4.1 Functional Requirements	43

4.4.2 Non-Functional Requirements	43
4.5 Summary	44
5 System Analysis	46
5.1 Introduction	46
5.2 Overview of System Architecture	46
5.3 Data Flow Analysis	47
5.4 Use Case Analysis	49
5.5 Entity-Relationship (ER) Analysis	50
5.6 System Constraints and Assumptions	52
5.7 System Use Case Diagram	53
5.8 System Use Case Description	54
5.9 System Class Diagram	56
5.10 System Interaction Diagram	57
5.11 System Activity Diagram	58
5.12 Summary	60
6.References	61
6 System Design	25
6.1 Overview of Design Approach	25
6.2 System Architecture Design	26
6.3 Database Design (if required)	27
6.4 User Interface (UI) Design	27
6.5 System Security Design (if required)	28
6.6 API Design (if required)	29
6.7 Integration Design	30
6.8 Summary	30
7 Implementation	32
7.1 Implementation Plan	32
7.2 Development of Core Modules	33
7.3 Integration of System Components	33
7.4 User Interface (UI) Development	34
7.5 Backend Development	35

7.6 Testing and Debugging	35
7.7 Deployment	36
7.8 Summary	37
8 Testing and Evaluation	39
8.1 Testing Strategy	39
8.2 Unit Testing	39
8.3 Integration Testing	40
8.4 System Testing	41
8.5 Evaluation of System Performance	42
8.6 Summary of Testing and Evaluation	42
9 Conclusion	45
9.1 Summary of Project Objectives	45
9.2 Key Findings and Results	45
9.3 Challenges and Solutions	46
9.4 Contribution to the Field	46
9.5 Recommendations for Future Work	46
10 References	48

2 List of Figures

Figure 1: Project TimeLine	23
Figure 2: Brain MRI sample from dataset	25
Figure 3: Brain Tumor Classification System Architecture	27
Figure 4: Type of entity	33
Figure 5: Location of the entity	33
Figure 6: Does the facility have a specialized brain tumor department	34
Figure 7: Number of doctors or staff specialized in brain tumors	34
Figure 8: internal electronic platform that documents test results or imaging	34
Figure 9: technological systems used in diagnosis or patient management ..	35
Figure 10: Current system efficiency: Satisfaction leve	36
Figure 11: the main current challenges you face in diagnosing brain tumors ..	36
Figure 12: Patient feedback on diagnostic delays and result clarity	36
Figure 13: Collaboration with research institutes for AI	37
Figure 14: intelligent electronic system can help reduce the diagnostic time ..	37
Figure 15: AI can improve the accuracy of brain tumor diagnosis	38
Figure 16: Functions do you expect to be most effective in the system	38
Figure 17: Type of system is preferred by the facility	38
Figure 18: Entity's technical readiness for AI	39
Figure 19: Main obstacles that may hinder the implementation	39
Figure 20: The results of AI-based systems	39
Figure 21: System connection to Ministry of Health for follow-up:Preferred ..	40
Figure 22: Will the system reduce diagnostic workload	40
Figure 23: Can the intelligent system reduce human errors in diagnosis	41
Figure 24: Does the facility agree to pilot the proposed system internally ..	41
Figure 25: type of reports does the facility want the system to generate	41
Figure 26: Should the system fully support the Arabic language	42
Figure 27: How important is doctor-to-doctor communication	42
Figure 28: Is the intelligent system practical to implement	42
Figure 29: The main components of the system Architecture	47

Figure 30: Brain Tumors Data Flow Diagram (DFD)	48
Figure 31: Entity-Relationship Diagram (ERD)	51
Figure 32: System Use Case Diagram	54
Figure 33: Brain Tumors System Class Diagram	57
Figure 34: Brain Tumors System Interaction Diagram	58
Figure 35: Brain Tumor Classification System Activity Diagram	59

3 List of Tables

Table 1: The Dataset	14
Table 2: The Group of Studies	17
Table 3: Question and options surveys	31
Table 4: Upload MRI Image – Use Case Description	55
Table 5: Analyze MRI Image – Use Case Description	56
Table 6: Save/Export Report – Use Case Description	57
Table 7: View Analysis Results – Use Case Description	58

Chapter 1

Introduction

1. Overview of the Project
2. Problem Statement
3. Goal and Objectives
4. Expected Outcomes
5. Summary

1 Introduction

1.1 Overview of the Project

This project aims to develop a system based on Deep Learning techniques to classify the stages of brain tumors using (MRI) scans. The importance of this project stems from the need for intelligent tools that assist doctors in accurately determining the tumor stage, which facilitates the creation of effective treatment plans and ensures more reliable follow-up of the patient's condition. The system relies on open-source data to ensure transparency and reproducibility. The model will be trained and tested according to accuracy and reliability standards to avoid errors and guarantee scientifically precise results. The project belongs to the field of healthcare and artificial intelligence, and it aims to support medical decision-making and improve the quality of healthcare by automating the process of tumor stage classification and providing accurate information that assists in diagnosis and treatment.

1.2 Problem Statement

Glioblastoma (GBM) is one of the most common and aggressive malignant brain tumors in adults, accounting for nearly half of all malignant gliomas and 15 of primary brain tumors. This tumor is characterized by its high ability to infiltrate brain tissues, significant cellular heterogeneity, and poor disease prognosis. Although most patients undergo surgical resection followed by chemotherapy and radiotherapy, clinical outcomes remain limited, with an average survival rate of only 15 months and a five-year survival rate not exceeding 5%. Current methodologies mainly focus on detecting the presence or absence of tumors. However, they often overlook the critical aspect of stage classification, which is essential for determining the progression of the tumor and accurately guiding treatment plans. Therefore, accurate diagnosis and proper classification of tumor stages are crucial factors in improving clinical outcomes and personalizing treatment strategies for each patient. MRI serves as a fundamental tool in this field due to its high precision and non-invasive nature. Based on this, the project aims to develop an artificial intelligence model using deep learning techniques to accurately classify the different stages of brain tumors, thereby assisting doctors in making more effective treatment decisions [1].

1.3 Goal and Objectives

1. To design and implement a deep learning-based system capable of accurately classifying the stages of a glioma brain tumor using MRI images.
2. To provide a practical AI-based tool that assists healthcare professionals in making well informed treatment decisions, enhancing monitoring of tumor progression, and improving patient outcomes.
3. To demonstrate the potential of deep learning with open-source data in medical imaging and provide a foundation for future research and development in automated tumor analysis and AI-assisted diagnostics.

1.4 Expected Outcomes

This project is expected to result in the development of an artificial intelligence model capable of accurately and reliably classifying the stages of brain tumors GBM based on MRI scans. The proposed system will contribute to improving the quality of clinical diagnosis by reducing complete reliance on human evaluation and providing an objective and rapid analysis of medical images. It is also expected that the model will demonstrate high efficiency in distinguishing the different stages of the tumor, enabling doctors to accurately track the progression of the condition and develop more personalized treatment plans.

1.5 Summary

The project's significance in the healthcare industry and its emphasis on creating a deep learning-based system to categorize brain tumor stages using MRI images were highlighted in this chapter's summary. It discussed the medical difficulties associated with GBM and the shortcomings of the available diagnostic techniques, emphasizing the necessity of precise and automated stage categorization. The project's purpose to create a trustworthy AI model that aids in clinical decision-making and enhances treatment results was outlined in the aims and objectives. The anticipated outcomes demonstrated the system's capacity to improve diagnostic accuracy and further medical imaging analysis.

The following chapter, will review relevant literature, highlight knowledge gaps in previous research, and assess whether the suggested model can be implemented ,providing a solid foundation for the project's development.

Chapter 2

Literature Review

2.1 Introduction

2.2 Background Research

2.3 Gaps in Literature

2.4 Feasibility Study

2.5 Summary

2 Literature Review

2.1 Introduction

This chapter provides a structured scientific overview of gliomas, including their types, causes, and characteristics, given their status as one of the most common brain tumors. It also reviews previous studies that have addressed brain tumor detection and classification using MRI and artificial intelligence (AI) applications. Furthermore, the chapter explores recent advancements in deep learning models within this field, highlighting the key challenges and existing research gaps that this project aims to address. The chapter concludes with a brief feasibility assessment of the project, addressing potential practical limitations and proposed solutions to overcome them, thereby enhancing the reliability and applicability of the expected results.

2.2 Background Research

2.2.1 Glioma Overview

Gliomas are the most common primary brain tumors of the central nervous system (CNS). Gliomas are subdivided into astrocytoma, oligodendroglioma, or glioblastoma based on two molecular markers: isocitrate-dehydrogenase (IDH) and 1p/19q co-deletion, where 1p refers to the short arm of chromosome 1 and 19q refers to the long arm of chromosome 19. Gliomas are a diverse group of tumors originating from glial cells in the brain and spinal cord, representing the most common primary brain tumor within the CNS. Gliomas, particularly high-grade forms like glioblastomas, pose numerous complications that significantly impact patient outcomes and quality of life, including increased intracranial pressure, seizures, hydrocephalus, brain herniation, hemorrhage into the tumor, local spread, spinal metastases, and death. These tumors vary widely in their aggressiveness and prognosis, ranging from low-grade pilocytic astrocytomas to highly malignant glioblastomas. Advances in molecular and genetic research have led to improved diagnostic accuracy and the development of targeted therapies. Despite these advancements, the diffuse and infiltrative nature of high-grade gliomas poses ongoing challenges in treatment and management. In the past, diffuse gliomas were classified and subtyped based on histopathology, such as diffuse astrocytoma, oligodendrogliomas, or mixed gliomas/oligoastrocytomas. More recently, gliomas have been classified based on molecular and genetic markers, leading to more specific prognostic and therapeutic targets. Factors that can increase the risk of glioma include aging, as gliomas are most common in adults between ages 45 and 65 but can occur at any age, exposure to ionizing radiation such as radiation therapy used to treat cancer, and, rarely, a family

history of glioma. Researchers have not identified definitive preventive measures. Researchers at the University of California, San Francisco (UCSF-PDGM) found that using AI to analyze MRI images can accurately predict tumor type and its genetic makeup[38] [39].

2.2.2 Related Dataset

The MRI Dataset UCSF-PDGM is a sizable and standardized collection of preoperative MRI data that was created to aid in the fields of radiogenomics, segmentation, and glioma classification research. Prior to receiving treatment, 495 adult patients with World Health Organization (WHO) grade II–IV diffuse gliomas who had 3-Tesla MRI scans at UCSF between 2015 and 2021 are included. Each record includes 11 three-dimensional MRI sequences, such as arterial spin labeling (ASL) perfusion, diffusion-weighted imaging (DWI), susceptibility-weighted imaging (SWI), high-angular-resolution diffusion imaging (HARDI), T1-weighted (pre- and post-contrast), T2-weighted, and FLAIR.

Standardization and preprocessing of all imaging data, including coregistration to a single 3D space, skull stripping, artifact reduction, and manual quality evaluation, were done to guarantee excellent consistency and picture quality. A combination of automated deep learning models and skilled manual correction is used in each case to segment the tumor in detail, classifying the lesion into enhancing tumor, necrotic or non-enhancing core, and peritumoral FLAIR abnormalities. The dataset encompasses imaging as well as clinical and genetic data, such as the presence of 1p/19q codeletion, O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation, and isocitrate dehydrogenase (IDH) mutation, as well as the extent of surgical resection and overall survival information.[42]

Table 2.1: The Dataset

ID	Dataset name	Type	Summary	Reference
1	TCGA-GBM/CCGA	MRI	(IDH, MGMT biomarkers)	[1]
2	TCIA	MRI	Predict patient survival.	[2]
3	Br35H, Figshare	MRI	Detect brain tumors.	[3]
4	Private Data-DICOM	MRI	Segment brain tumors.	[4]
5	Multi-Omics Data	Multimodal	Combine data to predict survival.	[5]
6	General Image Dataset	MRI	Classify brain tumors.	[6]
7	(T2)/(FLAIR)	MRI	Classify glioma grades.	[7]
8	BraTS	MRI	Segment and classify tumors.	[8]

2.2.3 Related study

Previous studies have indicated an increased interest in using Deep Learning approaches to analyze gliomas, with significant variance in aims and data sources, underlining the need for further research in the field of direct stage classification. In the field of molecular prognostication, some research integrated radiomics features extracted from MRI with pathological data to predict Overall Survival (OS), demonstrating that multimodal data integration improves survival prediction accuracy compared to using a single data source alone [14] [26].

Subsequent work introduced more comprehensive approaches by integrating radiological, genomic, pathological, and clinical data within advanced deep learning fusion models, leading to improved risk stratification and survival prediction performance [15][20]. For tumor detection and general classification tasks, transfer learning approaches based on VGG16/19 and ResNet50V2 architectures were

employed on public datasets, achieving tumor detection accuracies exceeding 98% [16][30].

In molecular and genetic prediction, several studies utilized 3D Convolutional Neural Networks (3D CNNs) to analyze volumetric MRI data and successfully predict key genetic markers such as MGMT and IDH, while also enhancing initial tumor classification performance[24] [27][28]. In parallel, U-Net-based convolutional neural networks were widely adopted for precise tumor segmentation, achieving high Dice similarity scores across different tumor regions[43] [29].

Other frameworks focused on developing fully automated CNN-based systems that combined segmentation and classification stages, reducing the need for manual intervention while maintaining high accuracy [17] [31] [34]. Lightweight CNN architectures were also proposed to address computational efficiency, achieving grade classification accuracies above 95% and maintaining high performance with reduced model complexity [25][18] [3].

Between 2019 and 2025, the application of Artificial Intelligence (AI) and Deep Learning (DL) in brain tumor MRI analysis evolved rapidly from early radiomics-based and 3D CNN models to highly efficient diagnostic frameworks. Research trends progressively shifted toward transfer learning to improve accuracy with limited data [19], direct tumor localization using object detection models such as YOLOv7 [32], and hybrid deep learning approaches that enhanced overall classification accuracy [35].

More recent studies emphasized the use of advanced CNN architectures, including Xception, ResNet, and MobileNetV2, achieving classification accuracies exceeding 99% while maintaining computational efficiency. In addition, emerging research directions incorporated Explainable AI (XAI) techniques to improve model interpretability [36] and Federated Learning frameworks to enhance patient data privacy and secure collaborative learning [37]. Overall, this progression highlights the critical role of AI and deep learning in advancing neuro-radiological diagnostics through accurate, efficient, and privacy-aware brain tumor analysis.

In addition, the study by S. Lysdahlgaard focused on analyzing diffuse gliomas using advanced MRI techniques, radiomics methods, and artificial intelligence models. It relied on the UCSF-PDGM dataset, which included nearly 500 patients with WHO grade II, III, and IV diffuse gliomas. The study extracted a large number of radiomics features from multi-sequence MRI images and used neural network models and the XGBoost algorithm to predict tumor grades and survival prediction. The study demonstrated high accuracy in segmentation and strong performance in predicting WHO tumor grades, as well as identifying important biomarkers using SHAP analysis. The study also highlighted the growing importance of artificial intelligence and radiomics in improving neuroradiological diagnosis, noting the need for larger and more diverse datasets to improve clinical performance and patient outcomes in the future [47].

Table 2.2: The Group of Studies

ID	Algorithm	Reference	Results
1	Radiomics (MRI + Pathology)	[9]	AUROC = 0.726 (TCGA-GBM test set)
2	Deep Orthogonal Fusion (DOF)	[10]	C-index = 0.788
3	VGG16/19, ResNet50V2 (Transfer Learning)	[11]	Acc = Prec = Rec= F1-score = 100%
4	3D CNN	[12]	Acc:0.806,Prec:0.801,Rec:0.927,F1-score:0.859
5	U-Net CNN	[13]	Acc:0.9936, SEN:1.0000, SPE:0.9706, AUC:0.9684
6	CNN Framework	[14]	Acc = 99.24%, Prec = 0.99, Rec = 99.18%, F1 = 0.99
7	Lightweight CNN	[15]	Acc:96.56%, Prec:92.81%, Rec:99.92%, F1: 96%
8	Lightweight CNN	[16]	Acc = 98.44%
9	Radiomics (MRI + Pathology)	[17]	C-index = 0.725 ; 6-month OS: Acc = 0.682, Prec = 0.904, Rec = 0.685, F1 = 0.779
10	3D CNN	[18]	AUCs: 0.790 (IDHwt), 0.737 (IDHmut-intact), 0.820 (IDHmut-codel)
11	CNN	[19]	Prec = 0.78, Rec = 0.79, F1 = 0.88
12	Transfer Learning	[20]	Acc = 95.27%, Prec = 95.50%, Rec = 95.27%, F1 = 95.29%
13	DOF Model	[21]	C-index = 0.788 (multimodal DOF), outperforming best unimodal model (0.718)
14	U-Net	[22]	Whole Tumor:0.89,eTumor Core:0.85 ,Enhancing Tu- mor:0.82
15	VGG16/19, ResNet50V2	[23]	Acc = 94.93%, Prec = 94.68%, Rec = 94.58%, F1 $\approx$ 94.6%

16	CNN	[24]	Acc = 98.73%, Precision = 99%, Recall = 99%, F1 = 99% (Confusion Matrix reported)
17	YOLOv7	[25]	Accuracy: 99.5% ,Precision: 99.5% ,Recall (Sensitivity): 99.3% ,F1-score: 99.4%
18	AI Review	[26]	AI in glioma (42 studies)
19	CNN	[27]	Glioma grading (2015-2023)
20	Lightweight CNN	[28]	Dice = 87.89%, mIoU = 81.02%, HD95 = 6.21 (UCSF-PDGM)
21	Hybrid DL	[29]	Acc=97.80%, Prec=97.90%, Rec=97.80%, F1=97.80%
23	Xception, ResNet	[30]	C-index = 0.723
24	MobileNetV2	[31]	Accuracy: 93.59% ,Precision (avg): 93.77% ,Recall (avg): 93.59% ,F1-score (avg): 93.65%
25	Neural Network, XGBoost	[32]	Acc = 95.00%

2.3 Gaps in Literature

Most existing research on brain tumor detection faces several limitations that reduce its effectiveness and real-world applicability. One major gap is the lack of data diversity. Many studies rely on datasets collected from a single source or under ideal imaging conditions, which leads to class imbalance and poor generalization when models are applied to real clinical environments. This project addresses this limitation by applying data augmentation techniques during the training phase, such as rotation, flipping, and intensity variation, to increase data variability and improve the model’s ability to generalize, despite using a single clinically validated dataset (UCSF-PDGM).

Another significant gap is the limited interpretability of deep learning models. Although models such as Convolutional Neural Networks (CNN) and ResNet achieve high predictive performance, they are often treated as black-box systems, which reduces transparency and clinician trust. To overcome this issue, the proposed system integrates Explainable AI techniques, specifically Grad-CAM, to visualize the regions of MRI images that influence the model’s predictions, thereby enhancing interpretability and supporting medical decision-making.

A further gap in the literature relates to practical clinical deployment. Many high-performing models suffer from high computational complexity, making them unsuitable for routine clinical use and causing performance degradation under varying imaging conditions. This project addresses this challenge by evaluating multiple deep learning architectures, including CNN, ResNet, and the lightweight MobileNetV2, with the goal of selecting models that balance accuracy

and computational efficiency, enabling practical and scalable deployment in real clinical environments.

2.4 Feasibility Study

The Brain Tumor Classification project demonstrates the technical feasibility of implementing advanced deep learning architectures such as CNN, ResNet, and MobileNetV2, which have proven effective in medical imaging and brain tumor research. We will use the UCSF-PDGM dataset, which is a clinically proven set of preoperative MRI images for glioma classification, to train and test the model. These algorithms are often used in neuroimaging because they can get high-level spatial features from MRI data. This guarantees that tumors are classified accurately and dependably in actual clinical environments.

2.5 Summary

This chapter presents a comprehensive scientific framework for studying gliomas, focusing on their clinical characteristics and the role of genetic markers, such as IDH and MGMT, in improving diagnostic accuracy and guiding treatment plans based on magnetic resonance MRI. The chapter also reviews recent advancements in employing artificial intelligence and deep learning techniques for brain image analysis, along with a review of relevant datasets, particularly the UCSF-PDGM dataset.

The review of previous studies highlights significant progress in the accuracy of deep learning models, but also acknowledges several challenges, including limited data diversity, high computational costs, and weak model interpretability and applicability in real-world clinical settings. The chapter also discusses the feasibility of the proposed project, emphasizing its potential implementation using efficient, lightweight models supported by data optimization techniques and interpretable artificial intelligence.

Based on the theoretical and analytical foundation presented in this chapter, the following chapter moves on to presenting the methodology used in implementing the project, where the structure of the proposed model, the stages of data pre-processing, the algorithms used, and the performance evaluation mechanism will be explained, in order to ensure that the research objectives are achieved systematically and accurately.

Chapter 3

Methodology

3.1 Introduction

3.2 Development Approach

3.3 Tools and Technologies

3.4 Project Timeline

3.5 Risk Management

3.6 Quality Assurance

3.7 Algorithm Description

3.5 Summary

3 Methodology

3.1 Introduction

This chapter outlines the methodology used to develop the proposed brain tumor classification system based on deep learning techniques. It details the practical and organizational steps taken to ensure the accuracy and applicability of the results in research and clinical settings. The chapter focuses on explaining the system design process, the selection of appropriate tools and techniques, and the organization of implementation phases, thus integrating the theoretical and practical aspects of the project. Section 3.2 addresses the development approach adopted in project implementation, explaining the method used to ensure the system's flexibility, modifiability, and incremental improvement. Section 3.3 presents the software and technical tools and techniques used in building the model. Section 3.4 outlines the project timeline, detailing the distribution of tasks and the timeframes required to complete each component of the system. Section 3.5 discusses the risk management strategies adopted to mitigate potential problems. Section 3.6 explains the quality assurance mechanisms and evaluation procedures used to verify the model's efficiency and performance accuracy. Section 3.7 provides a detailed description of the algorithm used to classify MRI images.

3.2 Development Approach

The project follows the Agile Methodology, which provides flexibility and continuous iteration, during the design, development, and testing stages. The work is divided into short sprints, each of which focuses on a specific activity such as data preprocessing, model training, or performance evaluation. The results of each sprint are reviewed and tweaked before proceeding to the next stage. Because it allows for gradual model improvement and parameter tuning based on test results, this technique is perfect for Deep Learning and Artificial Intelligence applications. It guarantees increased accuracy and consistent performance throughout the development process.

3.3 Tools and Technologies

The project was developed using the Python programming language as the primary language due to its extensive support for artificial intelligence and deep

learning applications, in addition to its high flexibility and great compatibility with various libraries, and its widespread use in medical image analysis research. The Visual Studio Code development environment was used, which provides effective tools for debugging, extension support, and systematic project organization that contributes to developing a structured and scalable AI system. The design, training, and evaluation of convolutional neural networks (CNNs) such as CNN, ResNet, and MobileNetV2 were carried out. Additional libraries such as NumPy, Pandas, OpenCV, and Matplotlib were used for data processing, image handling, and visually presenting results to ensure accurate analysis of MRI images. The backend system logic was implemented. Using the Python language for data processing and running the model (Model Inference), while a local SQLite database was used to store patient data, MRI image data, and classification results, due to its lightweight nature, ease of integration with Python applications, and suitability for standalone desktop systems. In addition, the Git system was used for version control, with the aim of tracking development progress and efficiently managing code modifications.

3.4 Project Timeline

Project Timeline

TASK	week 1	week 2	week 3	week 4	week 5	week 6	week 7	week 8	week 9	week 10	week 11	week 12	week 13	week 14	week 15	week 16
Discuss project																
Proposal																
Literature Review																
Methodology																
Requirement Gathering																
System Analysis																
Final Review																
Presentation																

Figure 3.1: Project TimeLine

3.5 Risk Management

Successful implementation of the proposed brain tumor stage classification system requires carefully identifying potential risks and developing appropriate mitigation strategies. Technically, limited computational resources may affect

the efficiency of training deep learning models, especially when processing MRI images; therefore, the model architectures are optimized, batch sizes are adjusted as needed, and lightweight models such as MobileNetV2 are used to ensure stable performance that aligns with the capabilities of the local system. There is also a potential risk of overfitting or poor generalization of the model, which could affect the clinical reliability of the system; to mitigate this, data augmentation techniques are applied, the data is correctly split into training, validation, and test sets, and performance evaluation metrics such as accuracy, recall, and precision are used to ensure balanced and reliable results. Since the system deals with sensitive medical data, data security and confidentiality are of critical importance; therefore, patient identities are anonymized before processing, and access to the system is restricted through authentication mechanisms. A local database of the SQLite type is also used to reduce exposure to external security risks. From a project management perspective, some delays may occur due to the complexity of model tuning and repeated experiments, and this risk is managed by following the Agile methodology and dividing the work into organized phases and iterative development cycles to ensure continuous progress and adherence to the schedule. Finally, user acceptance may pose a challenge, as some healthcare professionals may be hesitant to fully rely on AI systems; to enhance trust and transparency, explainable AI techniques such as Grad-CAM have been integrated to highlight areas influencing the model’s decisions and support the clinical interpretability of results.

3.6 Quality Assurance

To make sure the project is of high quality, the following steps will be taken. To make sure the model’s predictions are reliable, accuracy tests will use metrics like accuracy, precision, and F1-score as standard ways to measure them.

The model will be tested in terms of how long it takes to process MRI images and how well it does so during performance tests. There will also be code reviews to make sure that best practices are followed and mistakes are kept to a minimum.

3.7 Dataset Description

The proposed dataset, is The University of California San Francisco Preoperative Diffuse Glioma MRI Dataset (UCSF-PDGM) because [36] it provides one of the most comprehensive and clinically validated MRI collections for deep learning research on gliomas. The dataset includes 501 adult patients diagnosed with World Health Organization (WHO) grade II–IV diffuse gliomas, who underwent preoperative 3-Tesla MRI scans at UCSF between 2015 and 2021. Each record

contains eleven standardized 3D MRI sequences, including T1-weighted (T1) pre- and post-contrast, T2-weighted (T2), Fluid-Attenuated Inversion Recovery (FLAIR), Diffusion-Weighted Imaging (DWI), Susceptibility-Weighted Imaging (SWI), Arterial Spin Labeling (ASL) perfusion, and High Angular Resolution Diffusion Imaging (HARDI), all of which were pre-processed to ensure uniformity and high quality. In addition to imaging, the dataset incorporates essential clinical and genomic data, including Isocitrate Dehydrogenase (IDH) mutation, O6-methylguanine-DNA methyltransferase (MGMT) promoter methylation, and 1p/19q codeletion status, along with surgical details and patient survival outcomes. The tumors in this dataset are divided into three main compartments: the enhancing tumor (active region), the necrotic core (dead tissue area), and the peritumoral FLAIR abnormality (edema surrounding the tumor), using a combination of automated segmentation algorithms and expert manual annotations. This comprehensive design ensures that UCSF-PDGM is a high-quality, multi-modal dataset ideal for reliable and reproducible AI-based research in glioma classification, segmentation, and radiogenomic prediction. Samples from the dataset are illustrated in Figure 3.2.

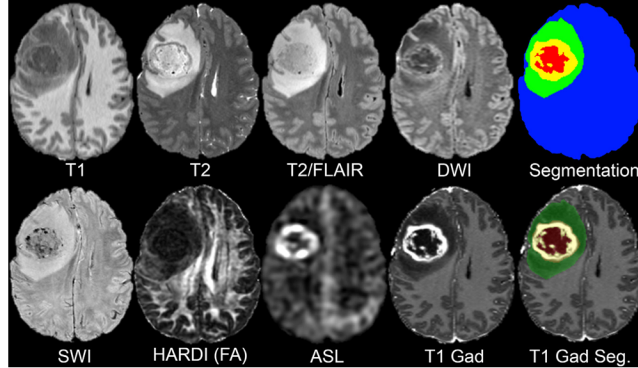


Figure 3.2: Brain MRI sample from dataset

3.8 Algorithm Description

This section provides a concise overview of the three algorithms employed in the brain tumor classification model. Section 3.8.1 describes the Convolutional Neural Network (CNN), which is used to extract spatial and structural patterns from MRI images with high precision. Section 3.8.2 presents the Residual Network (ResNet), which enhances model accuracy by incorporating skip connections that prevent information loss during training. Finally, Section 3.8.3 discusses MobileNetV2, a lightweight and efficient deep learning approach that ensures fast processing and reliable performance, making it well-suited for medical imaging applications.

3.8.1

The CNN is one of the most widely used and effective algorithms for analyzing MRI data. It operates through layers that automatically detect spatial and textural patterns in images without requiring manual feature extraction. It uses multiple layers that automatically learn patterns from MRI images, allowing the model to improve through hierarchical feature learning. CNNs are used to distinguish tumor regions from healthy tissues with high accuracy, making them ideal for brain tumor classification tasks. The algorithm relies on convolution and pooling operations to extract distinctive features at each layer, gradually improving model precision [37].

3.8.2

The ResNet represents an advanced evolution of deep neural networks, addressing the “vanishing gradient” problem that occurs in very deep models. It introduces skip connections that allow certain layers to be bypassed during training, enabling the network to learn complex image features without degradation in performance. ResNet is widely used in brain tumor classification because it maintains high accuracy even when trained on large-scale datasets such as UCSF-PDGM [40].

3.8.3

MobileNetV2 is a lightweight deep learning architecture designed for applications that require both speed and computational efficiency, particularly in medical systems with limited resources. It utilizes depthwise separable convolutions to significantly reduce computational cost while maintaining classification quality. In brain tumor classification projects, MobileNetV2 enables fast and accurate model execution, making it suitable for clinical deployment on portable devices or low-power environments [41].

3.8.4

In addition to using deep learning models for classification, an explainability approach was incorporated to enhance the interpretability of the system. While the custom CNN model achieved the highest classification accuracy, it did not produce clear visual explanations when applying Grad-CAM due to its relatively simple architecture.

Therefore, ResNet50 was utilized specifically for heatmap generation. Its deep convolutional layers provide richer feature representations, enabling more accurate localization of tumor regions in MRI images.

This approach represents a hybrid design, where the CNN model is used for tumor grade classification, and ResNet50 is used separately for visualization purposes. This combination allows the system to benefit from both high classification performance and effective visual interpretability.

3.9 System Architecture

This diagram illustrates the general workflow of the proposed system for analyzing and classifying brain tumors based on MRI images. The process begins with inputting image data from the UCSF-PDGM dataset, followed by a preprocessing stage that includes image registration, skull stripping, and value normalization to ensure consistent data quality. Next, the system performs tumor detection and segmentation to accurately locate the tumor within the MRI images. The processed data is then passed to a multitask deep learning model based on convolutional neural networks (CNNs), where analysis and classification tasks are carried out simultaneously. In the final stage, the system provides its outputs, including determining the tumor's genetic status based on IDH mutation status, as well as classifying the tumor grade according to the WHO into grade II, III, or IV as described in Chapter 5 section 5.2. The system architecture diagram is shown in Figure 3.3.

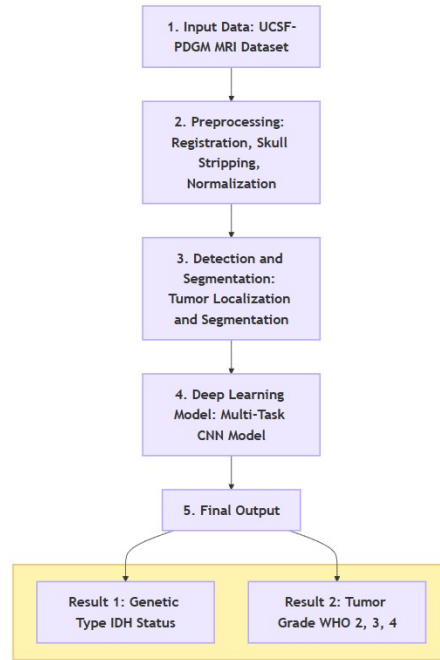


Figure 3.3: Brain Tumor Classification System Architecture

3.10 Summary

This chapter describes the steps used to create a brain tumor staging system utilizing deep learning algorithms. It starts by establishing the development methodology (Agile) and breaking down the work into short, sequential stages to guarantee continual model improvement. The chapter then introduces the tools and methods utilized, including Python, TensorFlow, and Google Colab, as well as a detailed timetable for the implementation steps. It also discusses risk management in terms of data quality, computing resources, and model correctness, outlining quality assurance techniques using suitable testing and assessment criteria. The chapter finishes by introducing the algorithms employed, such as CNN, ResNet, and MobileNetV2, and discussing their selection and function in performance improvement. Building on this basis, the chapter transitions to Chapter 4, which focuses on requirements gathering. This includes identifying stakeholders, deciding the required data gathering techniques, and developing the system's functional and non-functional requirements to assure the construction of a model that fulfills real-world clinical and applied demands.

Chapter 4

Requirement Gathering

4.1 Introduction

4.2 Stakeholder Identification

4.3 Methods of Requirements Gathering

4.4 System Requirements

4.5 Summary

4 Requirement Gathering

4.1 Introduction

The requirements gathering phase represents a critical foundation for project success as it lays the methodological foundation for transforming the general objective. (brain tumor classification) into detailed technical specifications. Due to the sensitive and complex nature of the proposed system, which uses deep learning models in a clinical environment, this chapter focuses on ensuring clinical reliability and operational efficiency. To achieve this, we start by identifying stakeholders, aiming to understand the expectations and needs of key parties, from doctors who seek high accuracy and interpretability of AI results, to hospital administrators who emphasize data security. Once these expectations are identified, we can move on to developing effective and reliable methods for gathering requirements. Finally, the Chapter discusses how to elicit and document system requirements, distinguishing between functional requirements that describe the core operations of the system (Such as automatic classificati) and the non-functional requirements that ensure high performance and accuracy in practical use.

4.2 Stakeholder Identification

- Doctors and radiology experts: they are the primary users who need a fast and accurate tool to classify brain tumors (grade II/III/IV) from (MRI) images. The focus is on clinical accuracy, ease of interpreting results (interpretable AI), and processing speed.
- Project development team: Responsible for building, training, and testing the deep learning model, in addition to developing the interface system. It focuses on meeting technical requirements, applying best programming practices, and ensuring code quality.
- Hospital management: focuses on improving patient outcomes, enhancing diagnostic efficiency, ensuring the safety of medical data, and ensuring the system complies with health regulations.
- Artificial intelligence (AI) researchers: They may use the system or its outputs for research purposes, and to access performance data and metrics.
- Patients: they are the indirect beneficiaries. Attention must be paid to ensuring accurate diagnosis and timely access to the appropriate treatment.

4.3 Methods of Requirements Gathering

During the development of the brain tumor classification system project, the main method used for data collection was a questionnaire. This independent approach helped us comprehensively assess the readiness of healthcare facilities to adopt an intelligent system. The questionnaire was created using Google Forms and included 25 questions. Its primary goal was to measure the technical and human readiness of healthcare facilities to adopt an AI-based system for brain tumor diagnosis, in addition to identifying current challenges and functional requirements. The questions covered several key areas: questions about the infrastructure and existing systems (such as HIS/PACS) to understand the operating environment, questions to assess the main challenges in the current manual diagnosis process (such as delays or lack of accuracy) and the level of satisfaction with current solutions, and critical questions to gather opinions regarding the adoption of the system and functional requirement preferences, including perceived effectiveness of AI, required features, potential obstacles (such as cost or data privacy), and a final assessment of the overall feasibility and practical applicability of the system within the healthcare environment. This structured approach ensured that system development aligned with the practical needs and technological capabilities of the target healthcare environment.

4.3.1 Analysis of survey responses

We analyzed 25 completed responses collected from healthcare facilities that participated in our survey. The reason for choosing this approach is to convert the raw opinions of professionals into quantitative data for system validation purposes and feature prioritization. Analyzing the responses helped us understand the current state of the healthcare environment, particularly concerning the level of technological infrastructure, the main challenges facing the current manual diagnostic process such as (delays or lack of accuracy), and the level of satisfaction with existing solutions. It also helped us determine the degree of readiness and confidence in adopting an AI-based system, as well as prioritize the most desired functionalities, such as tumor grade prediction and integration with the PACS system, which were key inputs for designing the system's functional requirements.

Table 4.1: Question and options surveys

Questions	Options
Type of entity	Government hospital, Medical diagnostic center, Private hospital
Location of the entity	Located in a large city, Located in a rural area, Located in a medium-sized city
Does the facility have a specialized brain tumor department	Yes , No
Number of doctors or staff specialized in brain tumors	Less than 5, From 5 to 10, More than 10
Is there an internal electronic platform that documents test results or imaging scans	Yes , No
What are the current technological systems used in diagnosis or patient management	Medical Imaging System (PACS), Hospital Information System (HIS), Custom electronic system, No system available
How satisfied are you with the efficiency of the current system used in diagnosis and follow-up	Very satisfied, Partially satisfied , Not satisfied
What are the main current challenges you face in diagnosing brain tumors	Delayed results, Weak integration between systems, Low accuracy in analysis , Lack of specialized staff
Are there recurring patient comments regarding long diagnostic periods or unclear results	Yes , No
Does the facility collaborate with research institutions to develop artificial intelligence solutions	Yes , No
Do you think an intelligent electronic system can help reduce the diagnostic time	Yes , No
Do you think artificial intelligence can improve the accuracy of brain tumor diagnosis	Yes , No
Which functions do you expect to be most effective in the system	Imaging analysis, Tumor grading prediction, Patient data management, Alerts and automated follow-up

What type of system is preferred by the facility	Internal desktop application, Medical website, Mobile application
Does the entity have sufficient technical infrastructure to operate an artificial intelligence system	Yes , No
What are the main obstacles that may hinder the implementation of the system	High cost, Resistance to change, Need for staff training, Privacy and data issues
How confident are you in the results of AI-based systems	Very high, Medium, Low
Do you prefer the system to be connected to the Ministry of Health database for case follow-up	Yes , No
Do you expect the system to reduce the workload on doctors during the diagnostic stages	Yes , No
Can the intelligent system reduce human errors in diagnosis	Yes , No
Does the facility agree to pilot the proposed system internally	Yes , No
What type of reports does the facility want the system to generate	Patient statistics, Diagnostic accuracy, Performance indicators
Should the system fully support the Arabic language	Yes , No
How important is it to have a feature for communication between doctors through the platform	Very important, Somewhat important, Not important
Do you think the intelligent system is practically implementable in healthcare facility environments	Not sure

The statistical analysis of the survey data on responding medical facilities shows variation in geographic distribution, types of institutions, and the specialized services they provide. As shown in Figure 4.1 (Type of Entity), the majority of participants were from public hospitals, accounting for 66.7%, followed by private hospitals at 25%, indicating that the study largely relied on data from the main public and private sectors. Regarding Figure 4.2 (Entity Location), it was found that an overwhelming majority of these facilities are located in a major city, representing 95.8%, highlighting that the potential scope of system application is primarily concentrated in urban centers. As for Figure 4.3 (Does the facility have a specialized department for brain tumors), the results showed a close split, with 54.2% of facilities having a specialized brain tumor department compared to 45.8% that do not. Similarly, Figure 4.4 (Number of doctors or staff specialized in brain tumors) indicates that 33.3% have fewer than 5 specialists, 25% have more than 10, and 20.8% have between 5 to 10 specialists, reflecting a generally diverse but limited availability of Specialists. Finally, Figure 4.5 (Internal Electronic Platform) shows that 83.3% of establishments use an internal electronic platform to document test or imaging results, indicating a high likelihood of seamless system integration.

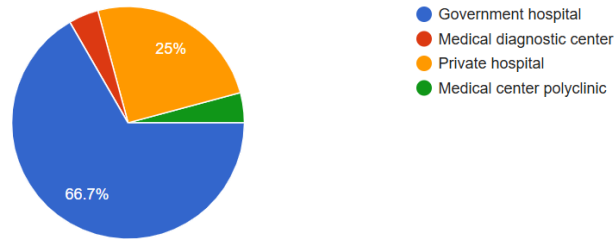


Figure 4.1: Type of entity

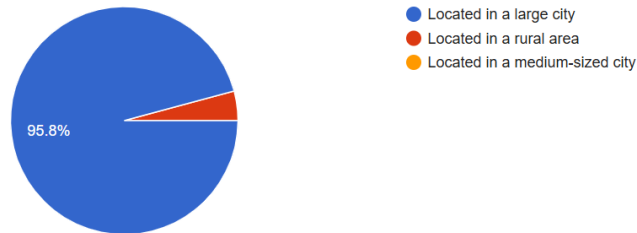


Figure 4.2: Location of the entity

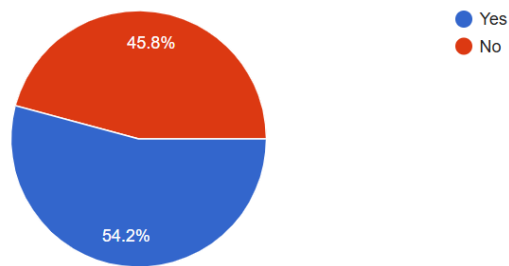


Figure 4.3: Does the facility have a specialized brain tumor department

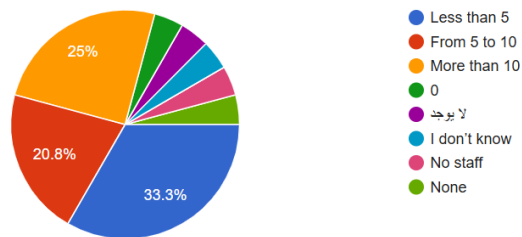


Figure 4.4: Number of doctors or staff specialized in brain tumors

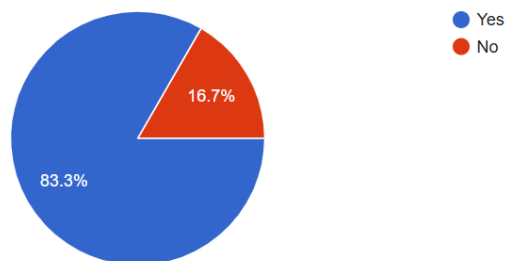


Figure 4.5: internal electronic platform that documents test results or imaging

As illustrated in Figure 4.6 which describes current Technological Systems Used in Diagnosis or Patient Management the indicated the use of Hospital Information

Systems (HIS) at 66.7% with Medical Imaging Systems (PACS) close, at 62.5%. Furthermore 29.2% depend on electronic systems whereas 4.2% stated they do not have any digital system. These findings emphasize that the majority of institutions are already functioning within a framework capable of supporting advanced diagnostic technologies. In Figure 4.7 (Satisfaction with the Efficiency of Current Systems) illustrates that satisfaction rates differ notably: 41.7% of participants indicated they were very satisfied 50% felt somewhat satisfied and 8.3% showed dissatisfaction. This pattern indicates that although numerous facilities appreciate their existing systems there is still potential, for enhancement particularly regarding diagnostic speed and precision. In Figure 4.8 (Main Current Challenges, in Diagnosing Brain Tumors) the primary problems consist of a shortage of personnel (50%) and slow turnaround times (41.7%). Poor system integration was indicated by 25% whereas 8.3% pointed to analytical precision. A minor portion referenced concerns or stated no difficulties. Together these challenges highlight functional deficiencies that an AI-powered diagnostic tool could address. In Figure 4.9 (Recurring Patient Comments on Long Diagnostic Periods or Unclear Results) 70.8% of participants reported that patients often express dissatisfaction with waiting times or ambiguous results. This significant percentage highlights a strong need, for more accurate diagnostic procedures. In Figure 4.10 (Collaboration with Research Institutions to Develop AI Solutions) illustrates a division, with 50% of facilities working together and 50% not engaging in collaboration. This suggests that although half of the organizations embrace innovation and research-based progress there is still room for expanding partnerships, in the future. In Figure 4.11 (Belief that an Intelligent Electronic System Can Reduce Diagnostic Time) reveals overwhelming support for AI implementation, with 95.8% of respondents agreeing that intelligent systems could shorten diagnostic time. Only 4.2% disagreed, strongly suggesting that the medical community perceives AI as a promising solution to existing inefficiencies.

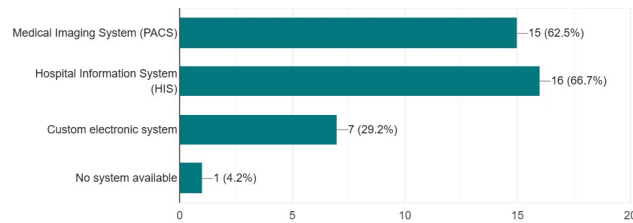


Figure 4.6: technological systems used in diagnosis or patient management

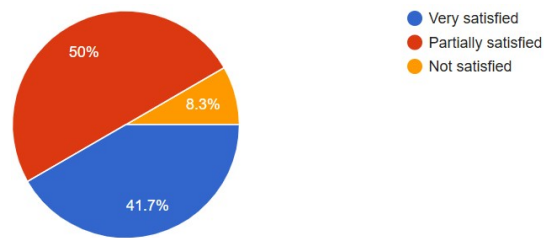


Figure 4.7: Current system efficiency: Satisfaction level

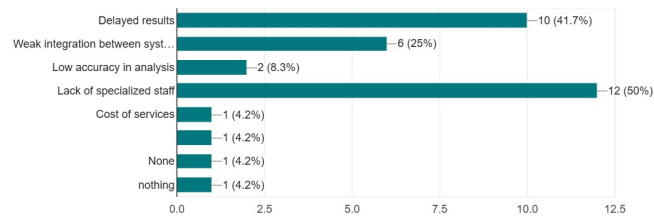


Figure 4.8: the main current challenges you face in diagnosing brain tumors

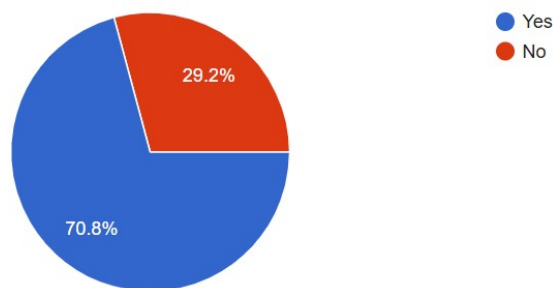


Figure 4.9: Patient feedback on diagnostic delays and result clarity

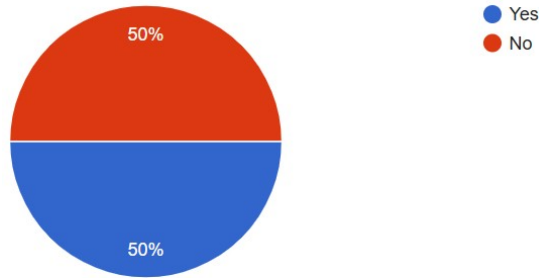


Figure 4.10: Collaboration with research institutes for AI

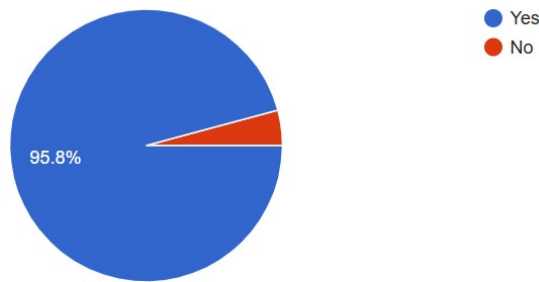


Figure 4.11: intelligent electronic system can help reduce the diagnostic time

In Figure 4.12, 100% of participants believe that artificial intelligence can improve the accuracy of brain tumor diagnosis. In Figure 16, image analysis was chosen as the most effective function by 75%, followed by tumor classification prediction at 54.2%, automated alerts and follow-up at 41.7%, patient data management at 29.2%, and finally, pathological results at 4.2%. In Figure 4.14, 50% of participants preferred an internal desktop application, 25% preferred the medical website, 20.8% preferred the mobile application, and 4.2% preferred the multidisciplinary system. In Figure 4.15, 54.2% of participants confirmed that they have sufficient technical infrastructure to operate the AI system, while 45.8% indicated that they do not have it. In Figure 4.16, 66.7% of participants identified high cost as the biggest obstacle, 54.2% pointed to the need for employee training, 41.7% mentioned resistance to change, 25% indicated privacy and data issues, 4.2% mentioned problems with the monitoring and maintenance team, 4.2% mentioned errors, and 4.2% said none. In Figure 4.17, 62.5% of participants showed moderate confidence in the results of AI-based systems, while 33.3%

showed very high confidence, and 4.2% showed low confidence. Finally, in Figure 4.18, 87.5% of participants supported linking the system to the Ministry of Health's database to follow up on cases, while 12.5% rejected this.

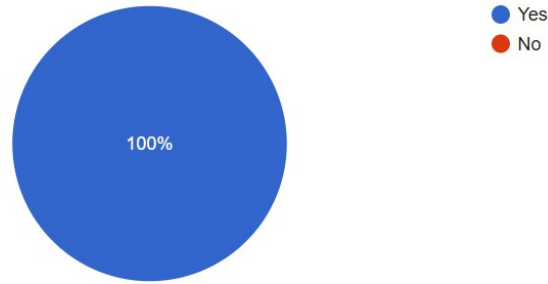


Figure 4.12: AI can improve the accuracy of brain tumor diagnosis

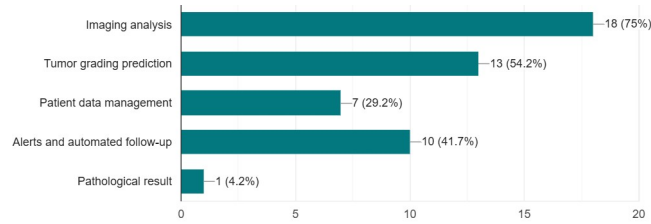


Figure 4.13: Functions do you expect to be most effective in the system

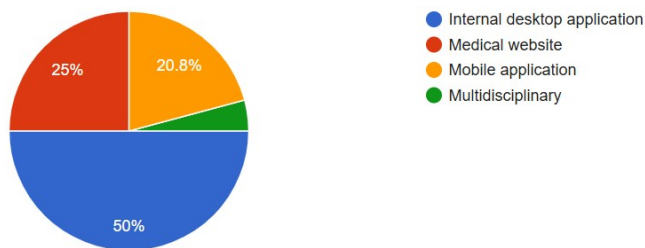


Figure 4.14: Type of system is preferred by the facility

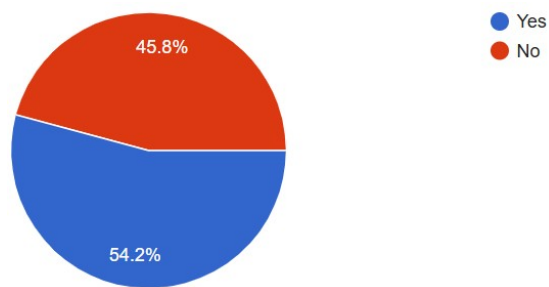


Figure 4.15: Entity's technical readiness for AI

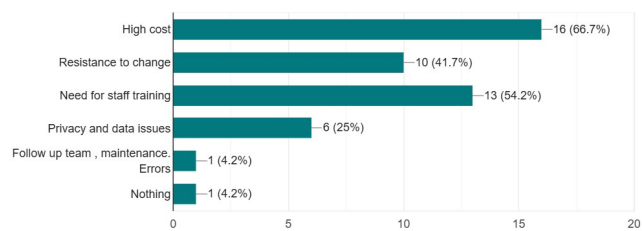


Figure 4.16: Main obstacles that may hinder the implementation

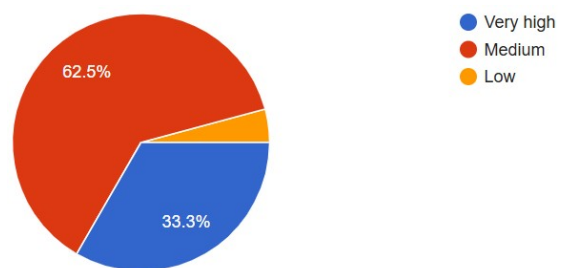


Figure 4.17: The results of AI-based systems

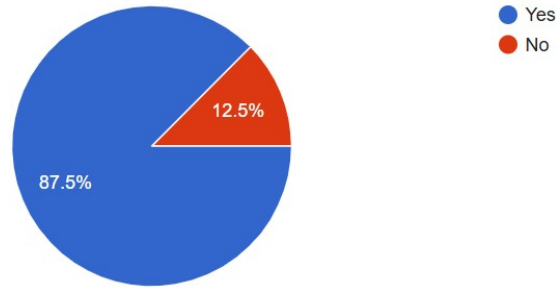


Figure 4.18: System connection to Ministry of Health for follow-up:Preferred

In Figure 4.19, 83.3% of participants expect the system to reduce the workload on doctors during the diagnostic phases, while 16.7% do not expect that. Also, in Figure 4.20, 79.2% of participants believe that the intelligent system can reduce human errors in diagnosis, while 20.8% believe it cannot. In Figure 4.21, 70.8% of facilities agree to internally trial the proposed system, while 29.2% refuse. In Figure 4.22, 75% of participants prefer the system to generate reports on diagnostic accuracy, 62.5% prefer reports on patient statistics, and 58.3% prefer reports on performance indicators. In Figure 4.23, 62.5% of participants want the system to fully support the Arabic language, while 37.5% do not. In Figure 4.24, 91.7% of participants consider it very important to have a feature for communication between doctors through the platform, while 8.3% consider it somewhat important. Finally, in Figure 4.25, 62.5% of participants believe that the intelligent system is practically applicable in healthcare environments, while 33.3% are uncertain, and 4.2% believe it is not applicable.

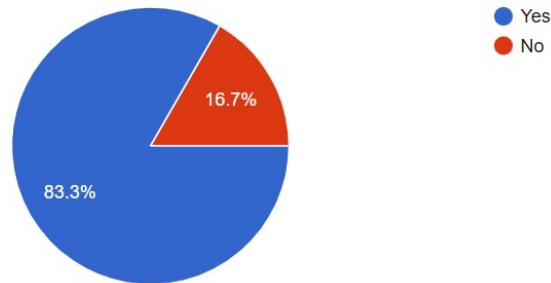


Figure 4.19: Will the system reduce diagnostic workload

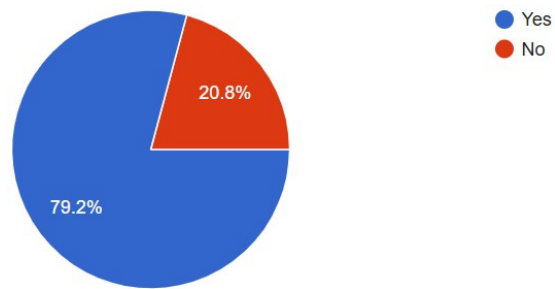


Figure 4.20: Can the intelligent system reduce human errors in diagnosis

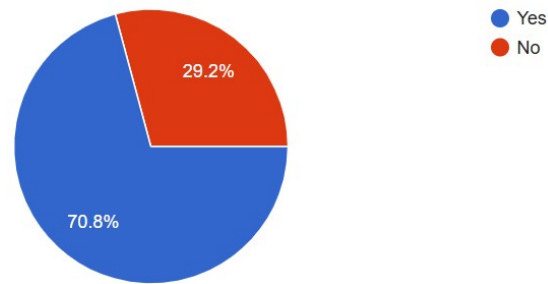


Figure 4.21: Does the facility agree to pilot the proposed system internally

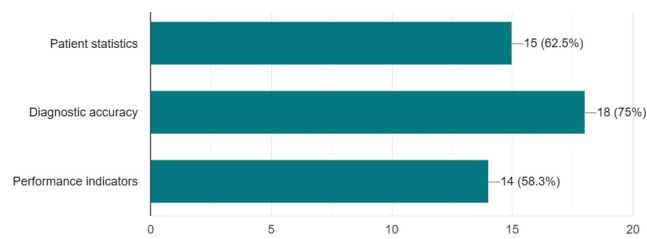


Figure 4.22: type of reports does the facility want the system to generate

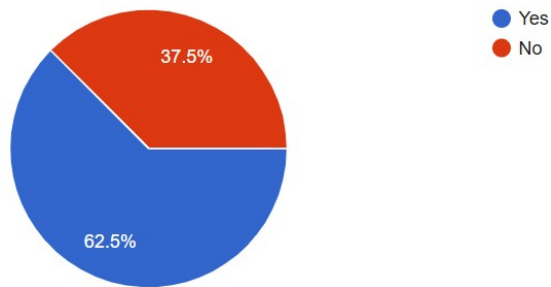


Figure 4.23: Should the system fully support the Arabic language

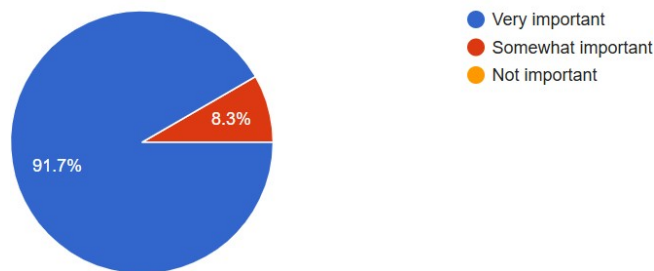


Figure 4.24: How important is doctor-to-doctor communication

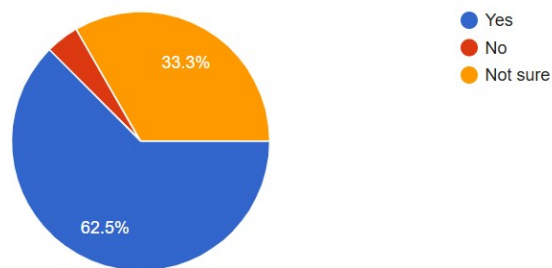


Figure 4.25: Is the intelligent system practical to implement

4.3.2 Analysis results

The survey collected provides a comprehensive view of functioning of medical institutions and their readiness to implement an intelligent diagnostic system.

Examination of the findings revealed that the majority of respondents are hospitals situated in metropolitan regions indicating that the suggested system would mainly cater to prominent healthcare facilities. The results also showed that certain centers possess brain tumor units while others lack them and that numerous organizations have a small pool of experts highlighting the necessity for technological assistance to address staffing shortages. Moreover the majority of centers depend on systems and existing frameworks for handling patient data indicating a solid base, for incorporating AI technologies. The study also identified operational issues, such as a lack of medical staff, slow diagnostic procedures and problems with system integration and precision all underscoring the necessity for enhanced diagnostic tools. Additionally participants noted that patients often voice worries about wait times and ambiguous outcomes highlighting the demand, for a more effective system. Regarding the role of AI, the survey showed strong confidence in its potential to enhance diagnostic accuracy, reduce human error, and support essential functions such as image analysis and automated follow-up. Based on these results, an intelligent diagnostic system can be designed to include advanced imaging tools, integrated reporting features, communication support between medical teams, and user-friendly functionalities tailored to healthcare environments.

4.4 System Requirements

4.4.1 Functional Requirements

The system requirements define the essential functions that must be provided to support the diagnosis of brain tumors and improve medical workflow within healthcare facilities.

1. **MRI Image Upload and Analysis** The system must allow for the uploading of MRI images for analysis using artificial intelligence algorithms to extract preliminary indicators for tumor staging.
2. **Results Visualisation and Reporting** The system should provide a clear visual display of the analysis results, allowing the physician to add comments or modify the report before approval.
3. **Patient Data Management** The system must support patient data management, linking current results to previous records.
4. **Alerts and Report Export** The system should also send alerts to the physician upon completion of the analysis or when the patient's condition is updated, with the ability to save or export reports via the internal medical platform.

4.4.2 Non-Functional Requirements

The non-functional requirements define the operational and quality standards that the system must meet to ensure reliable and efficient performance.

1. **performance** The system should provide a fast response time for image analysis and results display, ensuring smooth medical workflow and minimizing waiting times.
2. **Security** The system must also implement high security standards by storing patient data and sensitive medical information in an encrypted and reliable manner.
3. **Usability** The user interface is expected to be simple and clear, enabling physicians and technicians to use the system without difficulty.
4. **Integration** The system must be highly integrated with other hospital systems such as EMR and PACS, and operate efficiently as data volumes or the number of users increase within the healthcare environment.

4.5 Summary

The requirements for a brain tumor classification system have been successfully identified and analyzed. This phase began with stakeholder analysis, followed by a systematic survey whose results statistically demonstrated the need for an intelligent decision support system, particularly given the current diagnostic challenges and the availability of technological infrastructure in most healthcare facilities. This phase culminated in the formal documentation of eight functional requirements (FRs) and five non-functional requirements (NFRs), all clearly derived from and justified by stakeholder responses. This documented list of requirements now forms the foundation for the project's transition to the engineering phase. In Chapter 5 (System Analysis), these requirements will be analyzed in detail, the system architecture will be designed, and the specific software components and methodologies required to meet each requirement will be defined.

Chapter 5

System Analysis

5.1 Introduction

5.2 Overview of System Architecture

5.3 Data Flow Analysis

5.4 Use Case Analysis

5.5 Entity-Relationship (ER) Analysis

5.6 System Constraints and Assumptions

5.7 System Use Case Diagram

5.8 System Use Case Description

5.9 System Class Diagram

5.10 System Interaction Diagram

5.11 System Activity Diagram

5.12 Summary

5 System Analysis

5.1 Introduction

System analysis is a fundamental step toward developing an effective system, as it involves studying the main components of the system and analyzing how they interact. In this chapter, we start by providing an overview of the system's structure and explaining how data flows between the different components. We then address the Entity-Relationship (ER) analysis that represents the fundamental structure of the database. Due to the complex nature of the system, which relies on deep learning models to classify brain tumors, ensuring clinical reliability and operational efficiency is crucial. Therefore, the chapter begins with an analysis of data flow and use cases that define the system's main functions. After that, we present the class diagram and interaction diagram, which illustrate how the system components interact with each other.

5.2 Overview of System Architecture

The system relies on a client-server model where the user interacts with the user interface through the web. The AI-based server analyzes MRI images and classifies brain tumors.

The main components of the system is presented in figure 5.1:

- **Presentation Layer:** An interactive user interface through which doctors or specialists can add the patient's MRI images. Users can upload the images and view results such as type and stage of the tumor.
- **Business Logic Layer:** The processing is carried out using machine learning-based classification algorithms, which detect tumors in MRI images, determine the type of tumor (such as gliomas), and identify the stage (2, 3, or 4).
- **Data Layer:** MRI images and classification results are stored in a secure local database, with caching used to improve data retrieval speed and load images faster.

5.2.1 Architectural Pattern

Client Server

The client-server model was chosen because this pattern provides:

- **Flexibility:** Doctors can use the interface to upload and analyze data in real-time, while the AI-processing server performs the analysis and

classification.

- Scalability: Using servers allows for scaling to store and analyze large volumes of data.
- Security: Patient data can be securely stored on servers while ensuring information confidentiality.

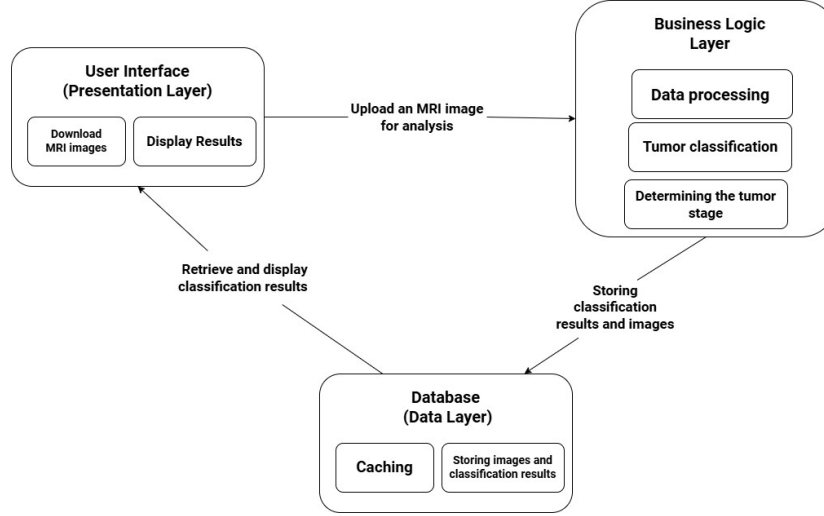


Figure 5.1: The main components of the system Architecture

5.3 Data Flow Analysis

The data flow begins when a radiologist or system user uploads an MRI image via the application interface. The system performs an initial quality check to ensure that the image format, size, and resolution meet the required standards, and only valid images proceed to the next stage. During preprocessing, the MRI image undergoes several enhancement operations such as skull stripping, noise removal, normalization, resizing, and contrast adjustment to ensure that the input image is consistent, clean, and ready for deep learning processing. Data augmentation is applied only during the model training phase when needed. After preprocessing, the refined image is passed to the selected deep learning model (CNN, ResNet, or MobileNetV2), which extracts relevant structural and texture-based features from the MRI scan and performs tumor grade classification. The output is one of the defined categories: Grade II, Grade III, or Grade IV, along with confidence scores that indicate prediction certainty. The classification results, including the predicted tumor grade, confidence values, and generated visualization maps

such as heatmaps (Grad-CAM), are temporarily stored in the system’s internal repository, allowing for result tracking, comparison, or exporting if required by the medical staff. Finally, the processed data is sent to the user interface, where the radiologist can view the tumor classification result, confidence percentage, and the highlighted tumor regions through visual explanation maps, and the user can then save, print, or export the report as needed.

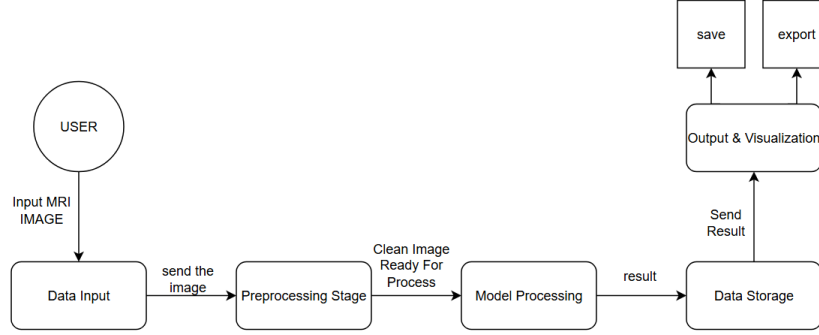


Figure 5.2: Brain Tumors Data Flow Diagram (DFD)

Figure 5.2 illustrates the DFD for brain tumor data. The diagram shows the basic steps the system goes through to process MRI images, from user data input to results display. These steps include image validation, preprocessing, deep model analysis, data storage, and finally, displaying results to the user.

1. User → Data Input Unit

- The interaction begins when the user uploads an MRI image via the system interface.
- The Data Input Unit receives this file and performs an initial check to verify the image format, resolution, and size to ensure they meet the minimum system requirements.
- Only valid images are sent to the next component.

2. Data Input Module → Preprocessing Stage

- Once the image passes the validation stage, the Data Input module sends the MRI image to the Preprocessing Stage. Here, several enhancement operations occur.
- These steps prepare a clear and uniform image, ensuring its optimization for efficient deep learning model inference.

3. Preprocessing Phase → Deep Learning Model Processing

- After preprocessing is complete, the resulting enhanced image is sent to the model processing unit. This unit loads the selected deep learning architecture (such as CNN, ResNet, or MobileNetV2)
- The model outputs a grade (II, III, or IV) along with confidence scores and visualization data where applicable.

4. Model Processing → Data Storage

- The classification output generated by the model is sent to the data storage unit.
- This step allows for later retrieval of results, tracking of patient history, and the ability to export data or generate reports.

5. Data Storage Unit → Output/Display Unit

- After the results are stored, the data storage unit sends the final processed data to the output/display unit.

6. Output & Visualization Module → User

- Finally, the system returns the processed output to the user interface.
- Print the data or export it using supported file formats.

5.4 Use Case Analysis

This analysis provides a systematic overview of the interaction between the physician and the system by identifying the core functions upon which the diagnostic workflow depends. The use cases directly reflect the functional requirements that the system must meet, demonstrating how key tasks such as uploading and analyzing MRI images are performed using artificial intelligence techniques. This analysis helps build a clear understanding of the role of each function within the system and defines its scope, paving the way for the development of the use case diagram in the next section and detailing the steps of each case in the subsequent executive description.

The system's 4 main use cases have been identified:

- Use Case 1: Uploading MRI Image
 - Actor: Physician
 - Steps: The physician selects the patient's MRI image from the machine and uploads it to the system. The system verifies the image format and quality, then saves the image and links it to the patient's record within the system.

- Outcome: The MRI image has been successfully uploaded and is available for analysis.
- Use Case 2: Analyze MRI Image
 - Actor: Physician
 - Steps: The physician selects to begin the analysis process after uploading the MRI image.

The system applies preprocessing to the image and then uses AI algorithms to extract initial indicators to determine the tumor stage.
 - Outcome: The MRI image has been analyzed, and tumor staging results have been generated.
- Use Case 3: View Analysis Results
 - Actor: Physician
 - Steps: The physician views the analysis results through the system interface. The system displays the tumor staging results clearly and visually, indicating the tumor stage and the confidence level of the result.
 - Outcome: The physician reviews the analysis results to support medical decision-making.
- Use Case 4: Save/Export Report
 - Actor: Physician
 - Steps: The physician saves the medical report within the system or exports it via the internal medical platform for use in medical procedures or archiving.
 - Outcome: The medical report has been successfully saved or exported.

5.5 Entity-Relationship (ER) Analysis

The ER model as shown in Figure 5.3 was designed to identify the fundamental logical components of a brain tumor classification system database. This model aims to ensure the efficient storage of all diagnostic data and medical records, making it a foundation for subsequent software design stages.

The brain tumor classification system includes major entities, which are:

- Physician: Represents the primary user of the system, responsible for reviewing diagnoses and overseeing patient records.
- Main attributes: PhysicianID (primary key), Name.

- Patient: Represents the medical records of the patient undergoing the scan, associated with the supervising physician.
- Main attributes: PatientID (primary key), PhysicianID (foreign key), Name, Age, Gender, RecordNumber.
- Scan: Represents the MRI scan record that is uploaded and processed by the model.
- Main attributes: ScanID (primary key), PatientID (foreign key), UploadDate, FilePath.
- DiagnosisResult: Represents the outputs of the AI model, which are the final results of the analysis.
- Main attributes: ResultID (primary key), ScanID (foreign key), TumorType, TumorGrade.

The main relationships within the system are clearly defined to ensure proper data integration; the relationship between the doctor and the patient is a one-to-many (1:N) relationship, where each doctor can oversee multiple patients, while each patient is overseen by only one doctor within the system. Similarly, the relationship between the patient and the scan is also a one-to-many (1:N) relationship, as each patient can undergo multiple scans over time, while each scan is associated with only one patient. Finally, the relationship between the scan and the diagnosis result is a one-to-one (1:1) relationship, as each scan provides a single, unique diagnosis result within the system, and each diagnosis result belongs to only one scan.

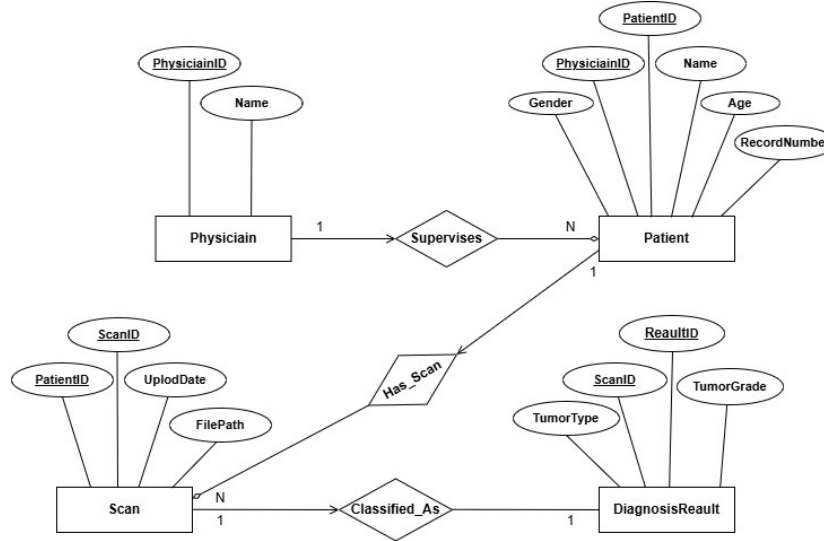


Figure 5.3: Entity-Relationship Diagram (ERD)

5.6 System Constraints and Assumptions

The system encounters restrictions due to the characteristics of MRI imaging the demands of deep learning and the technical settings, within healthcare institutions. These constraints were taken into account throughout system analysis and design to guarantee dependable operation.

- **Technical Limitations:** The classification model needs computational power to handle high-resolution MRI images. Insufficient GPU or hardware capabilities can lead to processing times and diminished responsiveness. differences in MRI machines, scan quality and imaging procedures between hospitals can impact prediction consistency. Heighten the demand, for preprocessing.
- **Data Accuracy and Reliability:** Model accuracy depends on the quality and completeness of MRI scans. Low-quality images, missing sequences, or noise can reduce classification performance. Since the model is trained on standardized open-source data, real clinical images may show variability that impacts prediction reliability.

5.6.1 Assumptions

- **User Behaviour:** It is assumed that users will upload clear, properly formatted MRI scans and interact with the system correctly to ensure meaningful outputs.
- **Technology Availability:** Healthcare facilities are assumed to have stable internet connectivity and devices capable of uploading and viewing MRI images through the system.
- **Data Integrity:** It is assumed that stored MRI data and system outputs follow validation rules that maintain accuracy, consistency, and secure handling of medical information.

5.6.2 Evaluation of Impact

- **Resource Constraints:** Due to computational limitations, essential functions such as preprocessing and tumor grade prediction must be prioritized, while more advanced features may require additional resources.
- **Technical Limitations:** Differences in MRI quality and scanning protocols may influence system accuracy. Additional preprocessing or validation may be necessary to maintain reliable performance across diverse clinical environments.

5.7 System Use Case Diagram

The use case diagram illustrates the various interactions between the primary user and the proposed system by representing a set of use cases that reflect the core functions supported by the system. This diagram aims to clarify the system's behavior and operational limitations from the user's perspective, and demonstrates how tasks related to brain tumor diagnosis and medical data management are performed.

The diagram, as shown in Figure 5.4, illustrates the following use cases:

- The system should allow the physician to upload the patient's MRI images for analysis.
- The system should support the analysis of MRI images using artificial intelligence algorithms to extract preliminary indicators for tumor staging.
- The system should display the analysis results in a clear and visual manner to support medical decision-making.
- The system should allow the saving or exporting of medical reports via the internal medical platform for use in medical procedures or archiving.

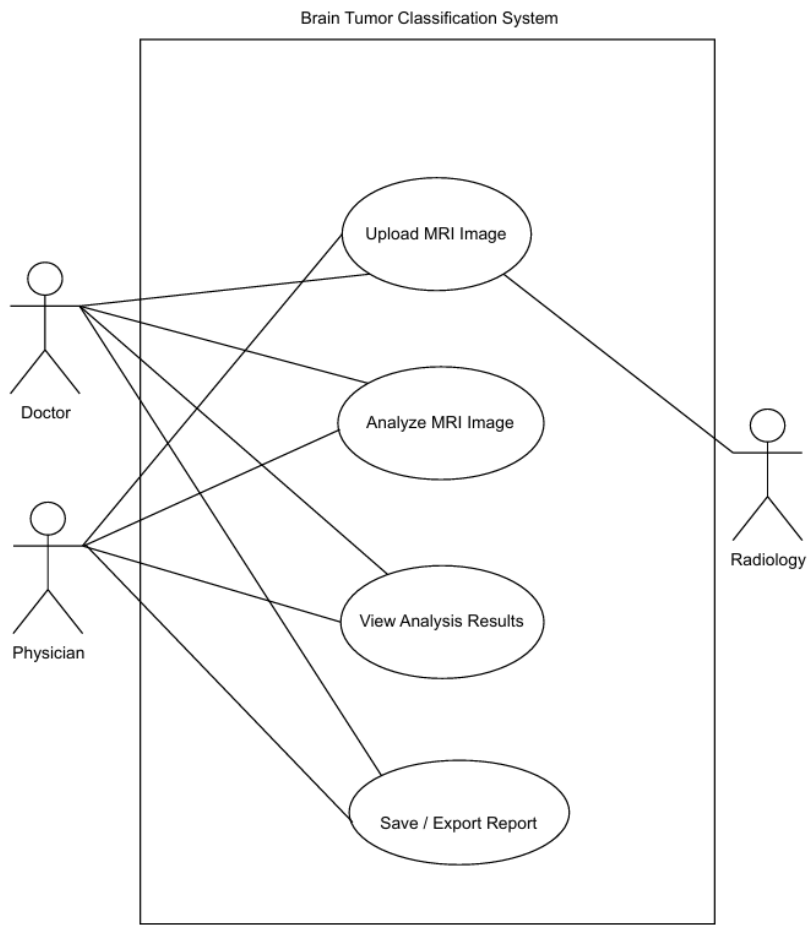


Figure 5.4: System Use Case Diagram

5.8 System Use Case Description

5.8.1 Upload MRI Image

Table 5.1: Upload MRI Image – Use Case Description

Use Case Name	Upload MRI Image
Primary Actor	Physician or Doctor or Radiology
Preconditions	The user must log in to the system and upload the MRI image in a supported format (DICOM, JPG, PNG).
Basic Flow	1. Upload MRI image via interface. 2. System receives, logs, and stores the file. 3. Image format and integrity are verified. 4. Image preprocessing is applied. 5. The system confirms the successful completion of the processing and begins analyzing it.
Post conditions	Image record is saved under the Scan entity and the image status is updated to Ready for Analysis
Alternative Flows	Invalid or corrupted images stop the process with an error message. Preprocessing failure is logged and the analysis preparation process is halted.

5.8.2 Analyze MRI Image

Table 5.2: Analyze MRI Image – Use Case Description

Use Case Name	Analyze MRI Image
Primary Actor	Physician or Doctor
Preconditions	The preprocessing of the MRI image has been completed and the images are ready for analysis.
Basic Flow	1. The system automatically detects MRI images ready for analysis. 2. Image is processed by CNN model. 3. The AI model, which uses a CNN algorithm, is used to process the image. 4. A CNN model analyzes the captured image of brain tumors by determining the type and grade of the tumor. 5. The analysis results are temporarily stored in the cache for quick access. 6. The system stores the final analysis results and processed images in the database.
Post conditions	DiagnosisResult is created with tumor status, stage, and confidence level.
Alternative Flows	Model failure or timeout results in a classification error for low-confidence results.

5.8.3 View Analysis Results

Table 5.3: View Analysis Results – Use Case Description

Use Case Name	View Analysis Results
Primary Actor	Physician or Doctor
Preconditions	The diagnosis result (DiagnosisResult) must be available in the system .
Basic Flow	1. The physician logs into the system. 2. The physician selects a patient or scan to view results. 3. The system retrieves the analysis results and processed MRI images. 4. The system displays the results, including tumor type, grade, and confidence level.
Post conditions	The physician successfully views the analysis results for the selected MRI scan.
Alternative Flows	If the results are not available or fail to load, the system displays an error message and logs the issue.

5.8.4 Save/Export Report

Table 5.4: Save/Export Report – Use Case Description

Use Case Name	Save/Export Report
Primary Actor	Physician or Doctor
Preconditions	The diagnosis result (DiagnosisResult) must be final and available for display.
Basic Flow	1. While displaying the analysis results, the operator clicks on the option to export or save the report. 2. It determines the output format, such as (PDF). 3. The system generates a report document that includes the MRI image and the final diagnostic classification (tumor type, grade).
Post conditions	A final and organized report document has been created and saved internally, or successfully exported.
Alternative Flows	If an error occurs during the report creation process, the system logs the failure and displays an error message.

5.9 System Class Diagram

The system class diagram in Figure 5.5 illustrates the main components of the system and how they interact to process MRI images and generate diagnostic results. The following provides a brief description of each class:

- **UIterface**: Represents the interface through which the user uploads MRI images, views results, and downloads reports. Connected to the **BusinessLogic** to initiate system operations.
- **BusinessLogic**: Coordinates the overall processing workflow. Linked to the preprocessing, model processing, and output components to complete the tumor analysis process.
- **PreprocessingStage**: Handles preparing MRI images for analysis through cleaning, normalization, noise removal, and resizing. Ensures image quality before model prediction.
- **ModelProcessingController**: Runs the classification model and predicts the tumor stage. Receives preprocessed images and produces the diagnostic result.
- **OutputController**: Responsible for displaying, saving, and exporting the generated reports. Retrieves stored results when needed.
- **DatabaseController**: Manages storing and retrieving scans, results, and patient data. Links the system's logic with the database entities.
- **Physician**: Represents the supervising physician with basic identification attributes. Connected to multiple patients.
- **Patient**: Contains patient information such as ID, age, gender, and medical record number. Linked to **Physician** and **Scan** entities.
- **Scan**: Represents an uploaded MRI scan with its metadata. Each scan is associated with one diagnosis result.
- **DiagnosisResult**: Stores the classification output such as tumor type, and grade. Linked directly to a specific scan.

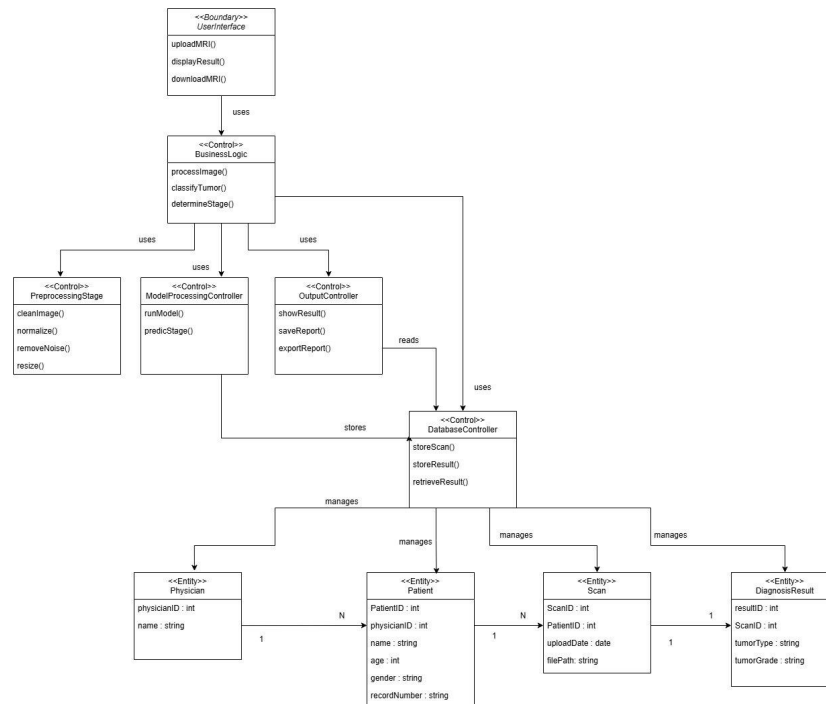


Figure 5.5: Brain Tumors System Class Diagram

5.10 System Interaction Diagram

The sequence diagram shown below 5.6 illustrates how different objects within the system interact in a specific time order to achieve a specific use case: MRI image analysis and diagnostic presentation.

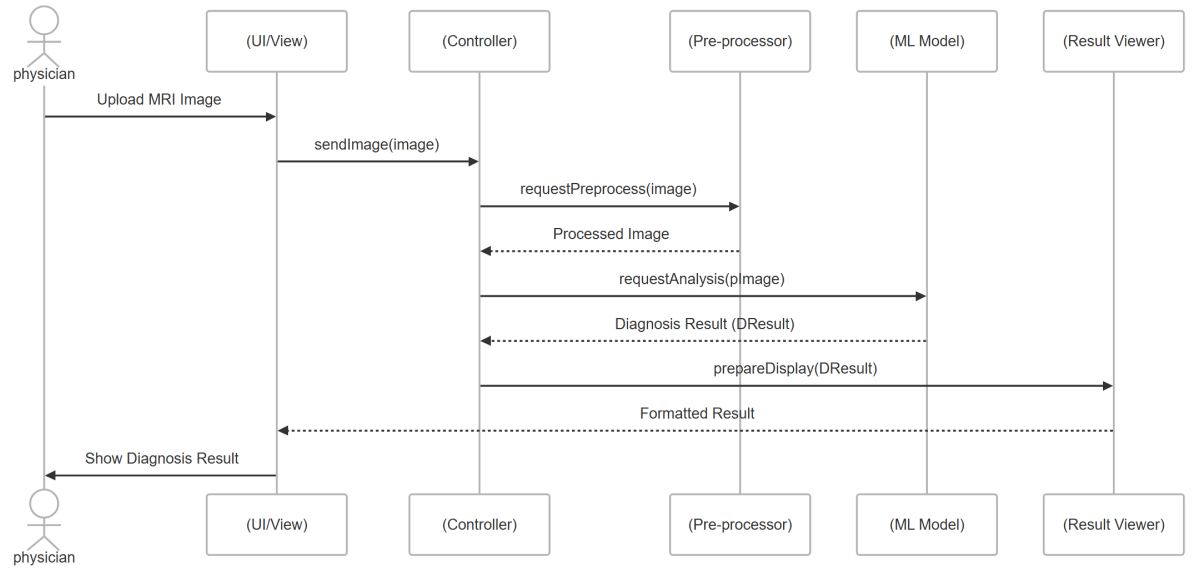


Figure 5.6: Brain Tumors System Interaction Diagram

5.11 System Activity Diagram

The activity diagram shows the logical flow of the steps and procedures carried out by the system during user interaction, detailing how the user completes various tasks, such as logging in and selecting a role, analyzing medical images MRI by the system, and providing classification results to the doctor and patient based on each user's permissions, as illustrated in Figure 5.7.

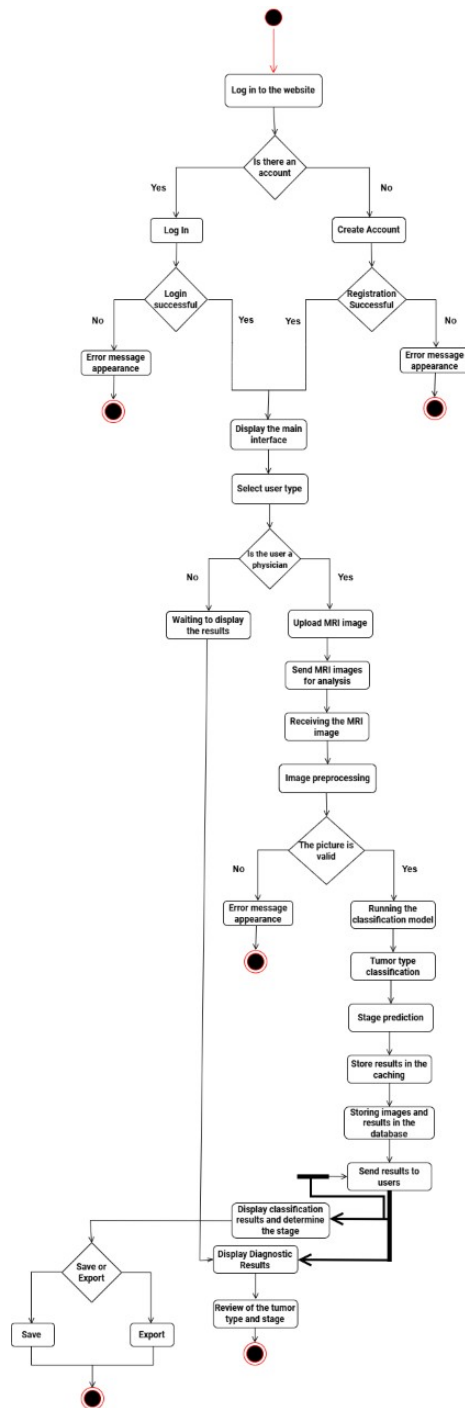


Figure 5.7: Brain Tumor Classification System Activity Diagram

5.12 Summary

In this chapter, the proposed system for classifying brain tumors using deep learning models was analyzed. The chapter began by providing an overview of the system's architecture, explaining how its various components interact, such as the user interface, business logic, and database, in addition to choosing a client-server model to ensure security and flexibility. The data flow was analyzed, from the user inputting MRI images, through the preprocessing stage, then analysis using artificial intelligence, and finally storing and displaying the results to the user. The main use cases of the system were identified, such as uploading images, tumor classification, and displaying results. An ER analysis was also presented to define the main entities in the system, such as the doctor, patient, MRI scan, and diagnostic result, and to clarify the relationships between them. Additionally, a class diagram and an interaction diagram were shown to represent the different components of the system and how they interact with each other. Finally, the functional and non-functional requirements were clarified to ensure classification accuracy and system performance in clinical healthcare environments.

Chapter 6

System Design

6.1 Overview of Design Approach

6.2 System Architecture Design

6.3 Database Design

6.4 User Interface (UI) Design

6.5 System Security Design

6.6 API Design

6.7 Integration Design

6.8 Summary

6 System Design

This chapter represents the pivotal stage where the theoretical analysis and requirements defined in previous chapters are transformed into a comprehensive design framework that clarifies the technical structure of the proposed system. Following the identification of the research problem and the analysis and development of the methodology, the system design phase translates these elements into a clear engineering diagram that defines how the system components will function and integrate practically.

System design is a strategic step that ensures the proposed system is feasible, flexible for future development, and capable of operating efficiently in a real clinical setting. Given the sensitive nature of AI-based medical applications, the design process involves not only defining the architectural structure but also considering safety, scalability, performance speed, ease of use, and ensuring the interpretability of results to support clinical decision-making.

This chapter explains the adopted design approach, the system architecture, database design, user interface design, security mechanisms, and API structure, aiming for integration with existing medical systems. This comprehensive presentation aims to build a solid technical foundation that ensures the system is ready for efficient and reliable implementation.

6.1 Overview of Design Approach

The proposed design methodology for the brain tumor staging classification system relies on developing a solution based on deep learning techniques characterized by accuracy, reliability, and applicability in clinical settings, for analyzing MRI.[20][29] The system was designed using a modular, multi-layered architecture to ensure clarity, ease of maintenance, and scalability. This methodology allows the system to be divided into independent components, including data preprocessing, model processing, result interpretation, and user interaction, enabling each part to be developed, tested, and improved independently.

The system relies on deep learning frameworks and tools such as Python and the TensorFlow and Keras frameworks to implement convolutional neural network models capable of extracting complex spatial features from MRI.[30][37] The overall design of the system focuses on compatibility with clinical environments by ensuring processing efficiency, high accuracy of results, and future scalability, such as integration with hospital systems (HIS/PACS) or adding other artificial intelligence models.

Key design principles include:

- **Modularity:** The system is divided into independent modules, including preprocessing of MRI images, tumor analysis and classification, presentation

of explanatory interpretations of the model, and delivery of results. This modular structure helps facilitate maintenance, updates, and replacement of any component without affecting the rest of the system.

- **Scalability:** The system is designed to handle an increasing volume of MRI image data and a growing number of users, by supporting optimization of computational resources and development of the models used. Additionally, using lightweight and efficient models such as MobileNetV2 contributes to scalability without putting a heavy computational load on the system.
- **Accuracy and Reliability:** High diagnostic accuracy is considered one of the fundamental principles in system design, achieved through relying on reliable, clinically validated datasets, and using several deep learning models such as CNN, ResNet, and MobileNetV2.[15][18][28] The system design also supports rigorous training, validation, and testing processes to ensure reliable results.
- **Security and Data Privacy:** Given the sensitivity of medical data, the system design focuses on privacy protection and information security. MRI images are handled after patient de-identification, and the system adheres to strict procedures for access control and secure data management in compliance with health data protection standards.
- **User-Centered Design (UCD):** The system was designed with the needs of medical staff in mind, focusing on ease of use, clarity of results, and interpretability. Explainable AI techniques, such as Gradient-weighted Class Activation Mapping (Grad-CAM), were also integrated to help doctors understand the model outputs and increase confidence in its decisions.[3][7]

The choice of this design methodology was based on the system requirements identified in previous chapters, which include the need for high diagnostic accuracy, interpretability of results, applicability in clinical environments, and performance efficiency. Constraints related to limited computational resources, data sensitivity, and the need for future scalability also influenced the selection of a modular, deep learning-based design architecture. This methodology ensures that the proposed system meets the current project objectives while providing a strong foundation for its future development and practical implementation.

6.2 System Architecture Design

The architecture of the proposed brain tumor staging classification system is based on a modular, layered architecture, where the system is designed as a (Web-based System) to support the analysis of MRI images using deep learning techniques.[20][29] The architecture consists of clear and interconnected components, including a web interface for user interaction, a backend web service to handle requests, and dedicated modules for preprocessing, model inference,

result interpretation, and report generation. This design ensures clarity, ease of maintenance, scalability, and compliance with the clinical requirements of the system.

key components including:

- **Web Interface:** It represents the primary point of interaction with the system, allowing doctors, radiologists, or physicians to upload MRI images, send analysis requests, and view tumor grade classification results (Grade II, III, IV) and reports.
- **Web Backend Service:** It represents the backend processing logic of the web-based system, where it receives requests from the web interface, validates the inputs, coordinates the analysis workflow by passing data to internal units, and then returns the final results to the interface.
- **Image Preprocessing Module:** Responsible for preparing MRI images before they are fed into the deep learning model, by performing operations such as resizing, normalization, and image quality enhancement, to ensure the consistency of input data and improve classification accuracy.[30][37]
- **Model Inference Module:** Runs the deep learning models used in the project (CNN, ResNet, MobileNetV2) to perform brain tumor stage classification and produce the final classification result.[15][18][28]
- **Explainability Module (Grad-CAM):** Provides a visual explanation of the model's decision by generating (Heatmaps), highlighting the areas the model relied on for classification decisions, thereby enhancing system transparency and doctors' confidence in its results.[3][7]
- **Report Generation Module:** Compiles analysis results, including tumor stage and visual explanation, into an organized report that can be displayed or saved by the user.
- **Database and File Storage:** It is used to store system data, such as analysis results, reports, and metadata, while ensuring data is managed securely and systematically to support future retrieval.

6.2.1 Architectural Diagram

Figure 6.1 illustrates the high-level component architecture of the web-based system for classifying brain tumor stages. The diagram shows the interaction between the web interface, the system's backend service, and the core processing modules, which include image preprocessing, inference using the deep learning model, result interpretation using Grad-CAM, and report generation. It also illustrates the flow of data within the system, from user input to storing and displaying the results.

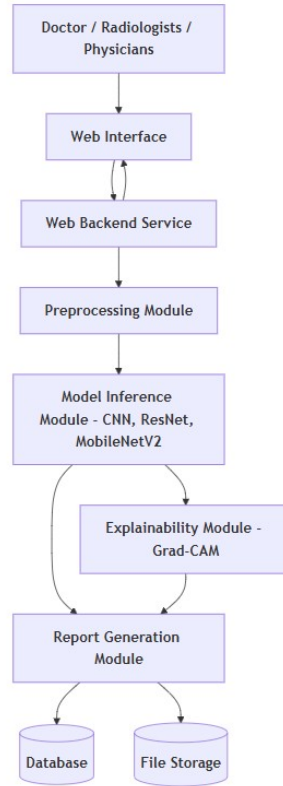


Figure 6.1: Component Diagram

6.2.2 Deployment Diagram

Figure 6.2 illustrates the system deployment architecture, showing how the system components are distributed between the client device (web), the system server, and the storage layer. The web interface runs on the user device, while the backend service and various processing modules run on the system server. The database and file storage are used to support the permanent and organized management of data, ensuring that analysis results and reports are preserved and can be referenced later.

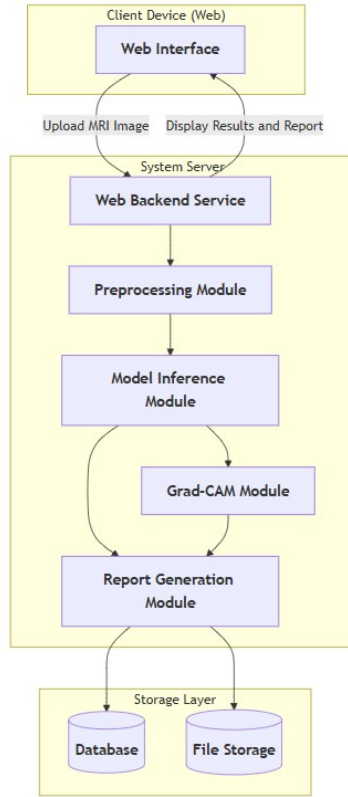


Figure 6.2: Deployment Diagram

6.3 Database Design

The database in this project has been designed to support a web-based brain tumor staging classification system using MRI images. The design relies on a Relational Database Model to ensure reliable storage and retrieval, tracking of uploaded MRI images and inference results generated by the AI model (tumor grades II / III / IV), in addition to supporting the storage of analysis outputs and their associated file paths such as (reports). The database schema consists of several interconnected tables, with each table representing a core entity within the system.

6.3.1 Database Tables

- User Table: Table 6.1 shows a table that stores information about authorized system users, such as (Doctors, Radiology specialists, and Physicians).

Table 6.1: User Table Structure

Field	Type	Key
UserID	INT	PK
Name	VARCHAR(100)	-

- Patient Table: Table 6.2 contains the basic patient data and is linked to the user table to identify the supervising user. Each patient can have multiple MRI images within the system.

Table 6.2: Patient Table Structure

Field	Type	Key
PatientID	INT	PK
Name	VARCHAR(100)	-
Age	INT	-
Gender	VARCHAR(10)	-
RecordNumber	VARCHAR(50)	UNIQUE
UserID	INT	FK

- Scans Table: Table 6.3 stores information about MRI scans, including upload details and file paths, and links each scan to its corresponding patient. It is the primary source for inputting data into the deep learning-based classification model.

Table 6.3: Scans Table Structure

Field	Type	Key
ScanID	INT	PK
PatientID	INT	FK
UploadDate	DATETIME	-
FilePath	VARCHAR(500)	-

- Diagnosis Results Table: Table 6.4 stores the classification results generated by the AI model, including the tumor grade (II / III / IV), with each diagnostic result linked to a single MRI image.

Table 6.4: Diagnosis Result Structure

Field	Type	Key
ResultID	INT	PK
ScanID	INT	FK
TumorGrade	VARCHAR(10)	-

6.3.2 Normalization

User data was stored in a separate Table User, patient data in a separate Table Patient, MRI scan data in a Scan table, and classification results in a Diagnosis Result Table. The tables are linked using primary and foreign keys, with each non-primary field directly dependent on its table's primary key, achieving the third normal form (3NF), reducing data redundancy, and ensuring data integrity.

6.3.3 ERD

Figure 6.3 illustrates the ERD of the system's database, showing the main entities: User, Patient, Scan, and Diagnosis Result, along with the relationships between them. The diagram indicates that a user can supervise multiple patients, that each patient may have multiple MRI scans, and that each scan is associated with a single diagnosis result representing the tumor grade (II / III / IV). This diagram reflects the logical design of the database and helps to understand the flow of data within the system and how the different components are linked to support medical image analysis and results storage.

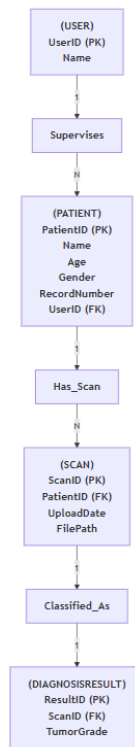


Figure 6.3: ERD of the system is database

6.4 User Interface (UI) Design

The User Interface (UI) of the proposed Brain Tumor Stage Classification System is designed to provide a clear, simple, and efficient interaction experience for

healthcare professionals. The interface follows a structured workflow that guides the user through authentication, MRI image submission, and result visualization.

The system consists of three main interface pages. Figure 6.5 shows the high-fidelity (Hi-Fi) UI design, while Figure 6.4 shows the low-fidelity (Lo-Fi) design of the system.

- **Login Page Interface:** The login page ensures secure access for authorized users. It contains input fields for the username and password, along with a login button to authenticate the user.
- **Main Dashboard (MRI Upload Interface):** The main dashboard represents the core operational interface of the system. It allows the user to select a patient by entering either the Patient ID or the Medical Record Number. The interface includes a dedicated button for uploading MRI images and a clearly defined image upload area that visually indicates where the MRI scans should be placed. Once the MRI images are uploaded, the user can initiate the analysis process using the “Start Analysis” button.
- **Tumor Analysis Result Interface:** The result interface displays the predicted tumor grade (Grade II, Grade III, or Grade IV). It also provides an option to download the diagnostic report in PDF format. The “Return to Dashboard” option allows users to navigate back to the dashboard.

Overall, the UI design emphasizes clarity, usability, and consistency. The interface is responsive and adapts to different screen sizes used in clinical environments while maintaining clarity and functionality. Accessibility is supported through readable text, clear color contrast, simple layouts, and clearly labeled components, helping reduce user effort and minimize interaction errors during clinical use.

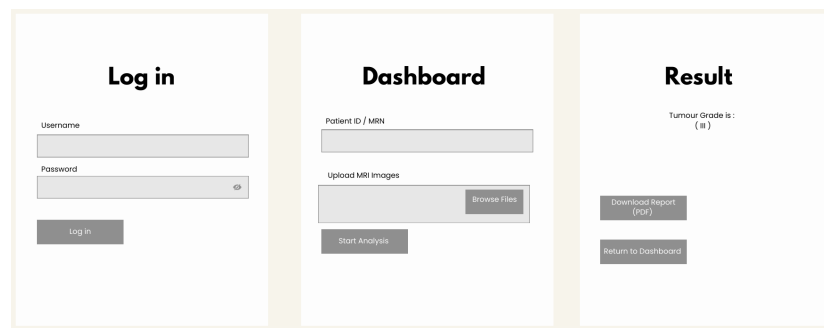


Figure 6.4: Low-Fidelity

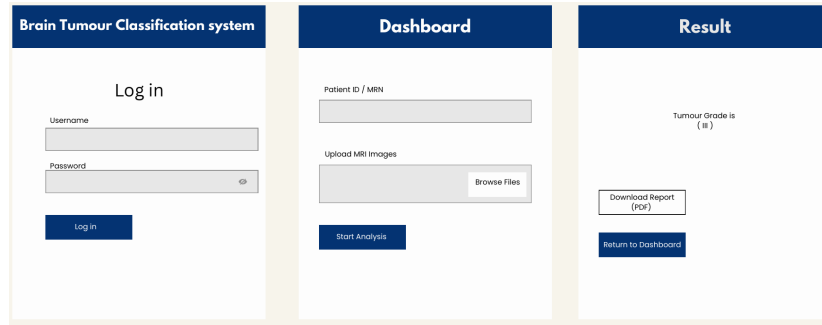


Figure 6.5: High- Fidelity

6.5 System Security Design

Security is priority for the brain tumor stage classification system, especially given the sensitive nature of the medical imaging data and patient records being handled. The system implements a robust security framework to ensure that only authorized medical personnel can access diagnostic results, protecting patient privacy and maintaining the integrity of the clinical data.

Key security measures include:

- **Authentication and Authorization:** The system employs secure login protocols to authenticate users. Role-Based Access Control (RBAC) is implemented to ensure that access to sensitive MRI data and classification reports is restricted to authorized doctors and specialists only.
- **Secure Password Storage (Hash + Salt):** User passwords are never stored in plain text. Instead, passwords are hashed using a strong cryptographic hashing algorithm (such as bcrypt or SHA-256) combined with a unique salt for each user. This approach protects against rainbow table attacks and significantly enhances resistance to brute-force attacks.
- **Data Encryption:** All medical images (MRI scans) and patient information are encrypted using advanced encryption standards (AES-256) both at rest in the database and in transit between the client interface and the backend server.
- **Data Integrity and Secure Communication:** To prevent unauthorized modifications of diagnostic results, HTTPS is enforced for all transmissions. Furthermore, the system ensures data integrity during the deep learning inference process, guaranteeing that the input MRI scan remains unaltered while being processed by the model.

6.6 API Design

The brain tumor stage classification application exposes several RESTful APIs to allow seamless communication between the user interface and the backend deep learning services. The API design follows RESTful principles, ensuring that each endpoint is stateless, scalable, and interacts with the appropriate classification and data services.

Key API endpoints include:

- POST /auth/login: Authenticates the medical personnel and returns a secure access token for session management.
- POST /classify/upload: Allows the user to upload an MRI scan to the server for processing.
- GET /results/scanId: Retrieves the classification results (Tumor Stage) and confidence scores for a specific uploaded scan.
- GET /history/patientId: Retrieves the previous classification records for a specific patient to monitor the disease progression over time.

The following Table 6.5 details the API endpoints, including the request methods, required parameters, and expected responses.

Table 6.5: ABI

Endpoint	Method	Parameters	Expected Response
/auth/login	POST	Username, Password	200 OK, Access Token
/classify/upload	POST	MRI Image File (PNG/JPG)	201 Created, Scan ID
/results/{scanId}	GET	Scan ID	200 OK, Tumor Stage (II, III, IV)
/history/{patientId}	GET	Patient ID	200 OK, List of past diagnoses

6.7 Integration Design

The Integration Design of the Brain Tumor Stage Classification System ensures seamless interaction between all system components and verifies that data flows correctly across the entire workflow. After successful user authentication through the login interface, the user is redirected to the main dashboard, where patient information is entered and MRI images are uploaded. Integration testing is performed to confirm that uploaded MRI images are correctly transferred from the user interface to the preprocessing module and subsequently to the CNN-based classification model.

The system verifies proper communication between modules by ensuring that image uploads reach the CNN model without data loss and that preprocessing steps are applied correctly. Once the CNN model generates the tumor grade prediction (Grade II, III, or IV), the results are returned to the user interface and displayed on the result page. The system also confirms that diagnostic reports are generated accurately and can be downloaded in PDF format.

During integration, potential challenges such as image upload failures, processing delays, data inconsistencies, and connection issues are addressed through validation checks, optimized processing pipelines, and error-handling mechanisms. These measures ensure that the system remains reliable, robust, and suitable for real-world clinical usage.

6.8 Summary

This chapter has established the comprehensive engineering framework for the proposed system for staging brain tumors using MRI images and deep learning techniques. The system's architecture, data flow mechanisms between its components, interface design, and security requirements have been explained. Design decisions have been linked to previously analyzed requirements to ensure consistency between theory and practice.

The adopted design aims not only to achieve accurate classification but also to provide a practical system characterized by reliability, maintainability, and scalability, while maintaining the security and protection of medical data. This design is a crucial step to ensure the system's applicability in real-world healthcare settings and its effective contribution to supporting medical decision-making and improving diagnostic quality.

Following the completion of the integrated design framework, the project moves to the implementation phase in the next chapter. This phase will review the development of software modules, the integration of system components, and the construction of the interface, ensuring the transformation of the theoretical design into a usable and functional system..

Chapter 7

Implementation

7.1 Implementation Plan

7.2 Development of Core Modules

7.3 Integration of System Components

7.4 User Interface (UI) Development

7.5 Backend Development

7.6 Deployment

7.7 Summary

7 Implementation

This chapter addresses the implementation phase of the proposed system for classifying brain tumors using deep learning techniques, following the completion of the analysis and design phases presented in previous chapters. The aim of this chapter is to translate the system's theoretical framework into a fully functional application capable of processing magnetic resonance imaging (MRI) images and delivering accurate and reliable results. The implementation process follows a structured methodology, beginning with a clear implementation plan, followed by the development of the system's core modules, and then the integration of its various components to ensure seamless operation. The chapter also focuses on developing a user interface (UI) that facilitates system use for clinicians, along with developing the backend responsible for data processing and running the deep learning model. Furthermore, the chapter reviews the testing and debugging processes to ensure the system's accuracy and stability. Finally, it addresses the system deployment phase and its preparation for use in real-world environments. Thus, this chapter forms a crucial link between the design phase and the final evaluation phase of the system.

7.1 Implementation Plan

The proposed brain tumor staging classification system was implemented using an Iterative Development Approach, where the system was developed and tested gradually to ensure reliability and the required functionality. The implementation process focused on building the core components of the system, integrating deep learning models to analyze MRI images, as well as developing a web interface that allows medical staff to interact with the system efficiently. The development process was carried out using Python as the primary programming language, with TensorFlow and Keras used to implement deep learning models, and Streamlit used to develop the web interface, while SQLite was used to manage the database. The implementation tasks were organized in several stages to ensure an organized and manageable development process.

The implementation plan was divided into three main phases:

- Phase 1: Development of the system's core units. This phase focused on developing the essential components required for the system. This included implementing the database structure using SQLite, developing user authentication and login functionalities, in addition to creating modules for managing patient information and uploading MRI images. The deep learning model for classifying brain tumor stages was also integrated, enabling the system to process MRI images and generate tumor grade predictions.
- Phase 2: Integration of system components and development of the user

interface. During this phase, the developed modules were integrated to ensure smooth communication between the web interface, backend system services, and the deep learning model. The user interface was implemented using Streamlit, allowing users to upload MRI images, initiate the analysis process, and display classification results. The interpretation module was also integrated using Grad-CAM to generate heatmaps highlighting important areas in the MRI images. Additionally, a function for generating diagnostic reports in PDF format was implemented.

- Phase 3: Testing, Debugging, and Preparing for System Deployment. The final stage involved fully testing the system to ensure that all components function correctly. This included verifying the image uploading functionality, checking the accuracy of tumor stage predictions, testing data storage and retrieval operations in the database, and ensuring that diagnostic reports are generated correctly. Programming errors and system issues were also addressed, in addition to performance improvements to ensure the system's reliability and ease of use.

main milestones include the following:

- Milestone 1: Completing the implementation of the user authentication system, setting up the database, and patient management modules.
- Milestone 2: Integrating the MRI image processing pipeline with the deep learning-based classification model.
- Milestone 3: Full system integration, testing the heatmap display module using Grad-CAM, the report generation module, and preparing for system deployment.

7.2 Development of Core Modules

The development of the core units of the brain tumor staging classification system focused on implementing the functional components specified in the system's architecture and design, as described in Chapter Six. The system was implemented using Python as the main programming language due to its strong support for medical image processing and deep learning frameworks.[37][30] Deep learning functions were implemented using TensorFlow and Keras[37], while the web interface was developed using Streamlit to provide an interactive environment for medical users. The system also uses SQLite as a lightweight relational database to manage user accounts, patient records, MRI images, and classification results.

The system consists of several basic units that work together to perform MRI image analysis and tumor stage classification[15][18]:

- **User Authentication and Account Management Unit** The user authentication unit has been implemented to ensure secure access to the system and to restrict its use to authorized medical personnel only. This unit manages login functions and verifies user data before allowing access to the system's control panel. User information is stored in an SQLite database and includes user IDs, usernames, and encrypted passwords. Password hashing techniques and secure credential verification have been implemented to protect user data. This module interacts directly with the database layer to verify login credentials and manage user sessions within the system. The system also supports role-based access control, where user roles such as administrator and standard user are stored in the database. The administrator role is identified through specific role attributes in the user table, allowing privileged access to system management functionalities, while regular users are limited to standard operations.
- In the implemented system, the login process is carried out by receiving the username and password from the user interface and validating them against the stored records in the SQLite database. The system checks whether the entered credentials match an existing user in the database before granting access to the dashboard. If the credentials are invalid, access is denied and an error message is displayed.

The login interface is implemented using Streamlit, where text input fields are used to collect the username and password from the user. When the login button is clicked, a verification function `verify_login` is called to validate the entered credentials against the stored records in the SQLite database. If the credentials are valid, the system saves the user information in the session state and redirects the user to the dashboard page using session management and rerun functionality. This ensures smooth navigation between pages within the application. However, if the entered credentials are incorrect, the system prevents access and displays an appropriate error message to inform the user.

The following figure 7.1 illustrates an example of the login implementation code in the system:

```

143 # ===== LOGIN PAGE =====
144 if st.session_state.page == "login":
145     # Display system title
146     st.markdown('<div class="center-bar">Brain Tumor Classification System</div>', unsafe_allow_html=True)
147     st.markdown('<div class="page-title">Log in</div>', unsafe_allow_html=True) # Display the page title
148
149     username = st.text_input("Username") # Username input field
150     password = st.text_input("Password", type="password") # Password input field
151
152     st.markdown('<div class="center-btn">', unsafe_allow_html=True)
153     if st.button("Log in"):
154         user = verify_login(username, password) # Calling the user data verification function
155         if user:
156             st.session_state.user = user
157             st.session_state.page = "dashboard" # Go to the dashboard page
158             st.rerun()
159         else:
160             st.error("Invalid Username or Password") # The data is incorrect, an error message appears
161     st.markdown('</div>', unsafe_allow_html=True)

```

Figure 7.1: Login page code

- **Database Management Unit** The database unit is responsible for storing and organizing all system data. The system uses SQLite as a relational database to manage information related to users, patients, MRI images, and diagnostic results. The database structure follows the schema described in Chapter Six and includes several tables such as User Table and Patient Table, Scan Table, Diagnosis Result Table These tables are interconnected using primary keys and foreign keys to maintain data consistency and enable efficient retrieval during system operation.
- **MRI Image Processing Unit** The MRI image processing unit prepares MRI images before sending them to the classification model[30][37]. MRI images are loaded from the uploaded scan files and processed using Python libraries such as NumPy, OpenCV, and Nibabel[37]. Since MRI images are stored as three-dimensional volumes composed of multiple slices, the system extracts the central slice from the scan for analysis[30]. Then the image is resized to 128×128 pixels to match the input size required for the trained convolutional neural network model. In addition, percentile normalization is applied to standardize pixel intensity values and improve the model's prediction performance. In the implemented code, the MRI scan is loaded using Nibabel and converted into a 3D numerical volume, from which the central slice is extracted for analysis. The extracted slice is then resized using OpenCV to match the required input dimensions of the model. Percentile-based normalization is applied using the 1st and 99th percentile values to enhance contrast and avoid the effect of outliers, while a small constant is added to prevent division by zero and np.clip is used to constrain values between 0 and 1. Finally, the grayscale image is converted into a 3-channel format and expanded into a batch dimension to make it compatible with the convolutional neural network input.

The following code snippet figure 7.2 illustrates a simplified example of the preprocessing process:

```

1 import tensorflow as tf
2 import numpy as np
3 import nibabel as nib
4 import cv2
5
6
7 model = tf.keras.models.load_model("models/best_model.h5")
8
9
10 #PREPROCESS
11 def preprocess_for_heatmap(file_path):
12
13     vol = nib.load(file_path).get_fdata()
14
15     z = vol.shape[2] // 2
16     slice_img = vol[:, :, z]
17
18
19     slice_img = cv2.resize(slice_img, (128, 128))
20
21     #normalization
22     p1, p99 = np.percentile(slice_img, 1), np.percentile(slice_img, 99)
23     slice_img = (slice_img - p1) / (p99 - p1 + 1e-8)
24     slice_img = np.clip(slice_img, 0, 1)
25
26     img = np.stack([slice_img]*3, axis=-1)
27     img = np.expand_dims(img, axis=0)
28
29     return img, slice_img

```

Figure 7.2: Preprocessing process

- **Tumor Classification Unit** The tumor classification unit performs the core task of the system, which is to predict the stage of the tumor[15][18]. The system uses a deep learning model trained and saved in the file `best_model.h5` to analyze MRI images and classify the tumor stage. The model is implemented using TensorFlow / Keras and relies on a deep convolutional neural network based on the ResNet50 architecture, which was trained on pre-labeled MRI image data[40][28]. After processing the MRI image, it is sent to the model for prediction, where the system classifies the tumor into one of the following grades Grade II, Grade III and Grade IV[15]. After that, the prediction result is saved in the database and displayed to the user. In the implemented code, the pre-trained model is loaded from the saved file using TensorFlow/Keras to ensure reuse without retraining. The model generates prediction probabilities for each class, which are then converted into the corresponding tumor grades using the defined labels dictionary. The final predicted class is obtained by selecting the highest probability value and mapping it to its medical stage representation for display and storage.

An example of the implementation:

```

import tensorflow as tf
import numpy as np

```

Loading the pre-trained model that was trained in the training phase

```

# The trained ResNet50 model
model = tf.keras.models.load_model("models/best_model.h5")

# Converting the model is digital outputs into tumor stages
labels = {
    0: "II",
    1: "III",
    2: "IV"
}

```

- **Model Selection** During the model development phase, several deep learning models were trained and evaluated, including CNN, MobileNetV2, and ResNet50[28][18]. Each model was trained using the same MRI dataset and evaluated based on performance metrics such as accuracy, precision, recall, and F1-score[15][18]. After comparing the results, the ResNet50 model achieved the best performance in terms of classification accuracy and generalization. Therefore, it was selected as the final model and integrated into the system for tumor classification.
- **Interpretation Unit Grad-CAM Display** To improve transparency in medical decision support systems, an interpretation unit was implemented using the Grad-CAM technique[3][7]. This unit generates heatmaps that highlight the areas in the MRI image that had the greatest influence on the classification decision made by the model[3]. The heatmap is generated from the convolutional feature maps of the neural network and then overlaid on the original MRI image. This visualization helps doctors better understand the reasoning behind the model's predictions.
- **User Interface Module** The user interface module was implemented using Streamlit, which allows for the rapid development of interactive web applications using the Python language. The Streamlit interface provides several features for the user, including Secure login and Patient selection, Uploading MRI images, Starting the tumor analysis process and Displaying classification results, Displaying heatmaps resulting from the interpretation module. This interface represents the main point of interaction between the user and the system's backend units. The login functionality is integrated within the Streamlit interface, where user inputs are captured and passed to the backend authentication module for verification. This ensures secure access control and seamless interaction between the user interface and the database layer. During the system implementation process, the team faced several development challenges. One of the most prominent challenges was dealing with large 3D MRI images and preparing them to be suitable for input into the deep learning model. This problem was solved by extracting the middle slice of the scan and resizing it to match the required input dimensions of the neural network. It also represents another challenge in providing a clear explanation of the model's predictions, which is particularly important in medical applications. This issue was addressed

by integrating the Grad-CAM technique to generate heatmaps that show the areas that influenced the classification decision.

Finally, ensuring smooth communication between the Streamlit interface, the database, and deep learning modules required careful system integration and testing. Several steps of debugging and result validation were carried out to ensure the reliability of the system's performance.

7.3 Integration of System Components

This section describes how the different components of the system were integrated to work together as a complete and functional application. The system combines the user interface, backend, database, and deep learning model into one unified workflow. The integration process begins when the user enters the patient ID and uploads an MRI image through the user interface. The system then communicates with the backend to validate the patient information from the database. After that, the uploaded image is processed and passed to the deep learning model for tumor grade prediction. Once the prediction is generated, the result is stored in the database and displayed to the user. This ensures seamless interaction between all system components.

Figure 7.2 This code represents the integration between the user interface, database, and deep learning model. It is executed when the user clicks the "Start Analysis" button. First, the system validates the input by checking if the patient ID is entered and exists in the database. If not, appropriate warning or error messages are displayed. It also ensures that an MRI image is uploaded before proceeding. Then, the uploaded file is saved locally, and if it is in NIfTI format (.nii or .nii.gz), a preview image is generated for visualization. After that, a scan record is created in the database to store the file information and link it to the patient. Finally, the deep learning model analyzes the MRI image to predict the tumor grade, and the result is saved in the database.

```

if st.button("Start Analysis"):

    if not record_number:
        st.warning("Enter Patient ID.")
        st.stop()

    patient = get_patient_by_record(record_number)

    if not patient:
        st.error("Patient not found in database.")
        st.stop()

    if not uploaded_file:
        st.warning("Upload MRI image.")
        st.stop()

    os.makedirs("uploads", exist_ok=True)
    file_path = os.path.join("uploads", uploaded_file.name)

    with open(file_path, "wb") as f:
        f.write(uploaded_file.getbuffer())

    # ===== NIFTI PREVIEW (PNG) =====
    st.session_state.preview_path = None
    lower_name = uploaded_file.name.lower()

    if lower_name.endswith(".nii") or lower_name.endswith(".nii.gz"):
        os.makedirs("previews", exist_ok=True)
        base_name = uploaded_file.name.replace(".nii.gz", "").replace(".nii", "")
        preview_path = os.path.join("previews", f"{base_name}_slice.png")
        nifti_to_png(file_path, preview_path, axis=2)
        st.session_state.preview_path = preview_path

    # scan
    scan_id = create_scan(
        patient_id=patient["PatientID"],
        file_path=file_path,
        original_file_name=uploaded_file.name
    )

    #Analysis
    grade, confidence = predict_tumor_grade(file_path)

    save_result(scan_id, grade)

```

Figure 7.3: System Integration Code

Figure 7.3 This code handles the preprocessing of MRI images in NIFTI format (.nii and .nii.gz). It first checks the file type, then creates a preview image by extracting a slice from the MRI scan and converting it into a PNG format. The preview is saved in a designated folder and stored in the session state, allowing it to be displayed in the user interface before performing the analysis.

```

uploaded_file = st.file_uploader("Upload MRI Image", type=["nii", "nii.gz", "jpg", "png", "jpeg"])
if lower_name.endswith(".nii") or lower_name.endswith(".nii.gz"):
    os.makedirs("previews", exist_ok=True)
    base_name = uploaded_file.name.replace(".nii.gz", "").replace(".nii", "")
    preview_path = os.path.join("previews", f"{base_name}_slice.png")
    nifti_to_png(file_path, preview_path, axis=2)
    st.session_state.preview_path = preview_path

```

Figure 7.4: Image Upload and Processing

Figure 7.4 This code calls the trained deep learning model to analyze the uploaded MRI image. The model processes the image and returns the predicted tumor grade along with a confidence score.

```

#Analysis
grade, confidence = predict_tumor_grade(file_path)

```

Figure 7.5: Model Integration

7.3.1 Testing of Integration

To ensure that the integrated system functions correctly, several testing strategies were applied. First, unit testing was conducted where each individual component, such as prediction functions and database operations, was tested separately to verify its correctness. Then, integration testing was performed to examine how different components of the system including the user interface, backend, machine learning model, and database work together and ensure proper data flow between them. Finally, system testing was carried out by simulating real user interactions, such as logging into the system, uploading MRI images, and viewing the analysis results. These testing strategies ensured that the system operates reliably and without errors. In the following section, we will present examples of errors as displayed in the system interface.

Figure 7.5 This interface displays a login error when incorrect username or password credentials are entered. The system ensures security by preventing unauthorized access and providing a clear error message to the user.

Log in

Username

ahmad

Password

...

Log in

Invalid Username or Password

Figure 7.6: Login Page

Figure 7.6 This interface shows an error message when a patient ID is entered incorrectly or does not exist in the database. The system validates the input and alerts the user with a clear message indicating that the patient was not found.

Patient ID / MRN

UCSF-PDGM

Upload MRI Image

Drag and drop file here

Limit 200MB per file • NII, NII.GZ, JPG, PNG, JPEG

Browse files

Start Analysis

Patient not found in database.

Figure 7.7: Patient ID Entry

Figure 7.7 This interface demonstrates a file upload error. The system rejects unsupported file formats and notifies the user that only specific image formats like MRI files are allowed.

Patient ID / MRN

UCSF-PDGM-016

Upload MRI Image

Drag and drop file here

Limit 200MB per file • NII, NII.GZ, JPG, PNG, JPEG

Browse files

application/vnd.openxmlformats-officedocument.wordprocessingml.document files are not allowed. ✕

Start Analysis

Figure 7.8: Unsupported File Upload

7.4 User Interface (UI) Development

The User Interface (UI) of the proposed Brain Tumour Classification System was developed using Streamlit, a Python-based framework that enables the creation of interactive web applications. The interface utilizes built-in Streamlit components such as buttons, input fields, and file uploaders to create a functional and efficient user experience. The UI was implemented based on the system design outlined in Chapter 6. The interface is simple and user-friendly, structured using Streamlit components to guide users from login to result display in a clear and organized manner. The interface is responsive and adapts to different screen sizes, ensuring usability across various devices.

The interface is structured into multiple sections using Streamlit layout features such as containers and columns. User inputs, including patient ID and MRI images, are captured and passed to the prediction model, and the results are displayed dynamically without requiring page reloads.

The following describes the main screens of the application:

- Login Screen: Figure 7.9 shows the login screen, which is the first screen displayed to the user. It includes fields for entering the username and password. A login button is provided to validate user credentials stored in the system database and allow secure access to the application. This screen was built using Streamlit input fields and a login button. The entered data is checked against the system to verify user credentials.
- As shown in Figure 7.10 the implementation code of the login screen. The `st.text_input()` method is used to create the username and password fields,

while `st.button()` creates the login button. When the button is clicked, the system validates the entered information using the `verify_login()` method. If the login is successful, `st.session_state` stores the user data and `st.rerun()` redirects the user to the dashboard page. Otherwise, `st.error()` displays an error message.

Figure 7.9: Screen of log in

```
if st.session_state.page == "login":

    st.markdown('<div class="center-bar">Brain Tumor Classification System</div>', unsafe_allow_html=True)
    st.markdown('<div class="page-title">Log in</div>', unsafe_allow_html=True)

    username = st.text_input("Username")
    password = st.text_input("Password", type="password")

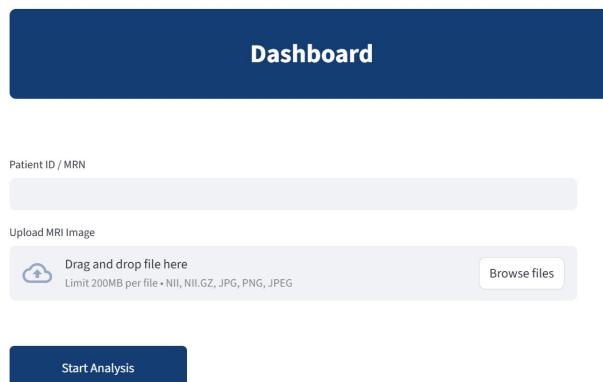
    st.markdown('<div class="center-btn">', unsafe_allow_html=True)
    if st.button("Log in"):
        user = verify_login(username, password)
        if user:
            st.session_state.user = user
            st.session_state.page = "dashboard"
            st.rerun()
        else:
            st.error("Invalid Username or Password")
    st.markdown('</div>', unsafe_allow_html=True)
```

Figure 7.10: Code of Log in Screen

- **Dashboard Screen:** Figure 7.11 shows the control panel screen that appears after successful login. The user is directed to the main interface. This screen allows the user to enter the patient ID and upload the MRI image. The entered patient ID is validated against the system database to ensure that the patient exists before proceeding with the analysis. The layout of this screen was designed using Streamlit containers and columns to

organize input elements clearly, while the file uploader component enables seamless image submission for processing.

- Figure 7.12 shows the implementation code of the dashboard screen. This section allows the user to enter the patient ID and upload the MRI image using `st.file_uploader()`. The system validates the entered data using `get_patient_by_record()` to verify whether the patient exists in the database. If required information is missing, `st.warning()` or `st.error()` messages are displayed, and `st.stop()` stops the process. After validation, `os.makedirs()`, `open()`, and `f.write()` are used to save the uploaded MRI image into the uploads folder.



The dashboard interface consists of a dark blue header with the word "Dashboard" in white. Below the header, there is a light gray input field labeled "Patient ID / MRN". Underneath this is another light gray section labeled "Upload MRI Image". Inside this section, there is a light blue box with a cloud icon and the text "Drag and drop file here" and "Limit 200MB per file • NII, NII.GZ, JPG, PNG, JPEG". To the right of this box is a button labeled "Browse files". At the bottom of the dashboard is a dark blue button labeled "Start Analysis".

Figure 7.11: Screen of dashboard

```

elif st.session_state.page == "dashboard":

    st.markdown('<div class="center-bar">Dashboard</div>', unsafe_allow_html=True)

    record_number = st.text_input("Patient ID / MRN")
    uploaded_file = st.file_uploader("Upload MRI Image", type=["nii", "nii.gz", "jpg", "png", "jpeg"])

    st.markdown('<div class="center-btn">', unsafe_allow_html=True)
    if st.button("Start Analysis"):

        if not record_number:
            st.warning("Enter Patient ID.")
            st.stop()

        patient = get_patient_by_record(record_number)

        if not patient:
            st.error("Patient not found in database.")
            st.stop()

        if not uploaded_file:
            st.warning("Upload MRI image.")
            st.stop()

        os.makedirs("uploads", exist_ok=True)
        file_path = os.path.join("uploads", uploaded_file.name)

        with open(file_path, "wb") as f:
            f.write(uploaded_file.getbuffer())

```

Figure 7.12: Code of Dashboard Screen

- **Result Screen:** Figure 7.13 shows the results screen that appears after the analysis is complete. The system displays the predicted tumour grade. An optional button is provided to download the result as a PDF report [Figure 9.1], which contains detailed patient information and classification. The report includes the patient ID, patient information (including gender, number of previous scans, and scan date), tumour classification, and the model confidence score. It also includes the MRI image with heatmap visualization to explain the model's decision-making process. This visualization highlights the specific areas in the MRI scan that the AI focused on to identify the tumour, making the classification results interpretable and verifiable by specialists. This screen dynamically presents results generated by the model, ensuring real-time feedback to the user through Streamlit display components.
- Figure 7.14 shows the implementation code of the result screen. This code displays the result page after the MRI analysis is completed. The system retrieves the prediction results using `st.session_state.get()`. If no result exists, `st.warning()` displays a warning message. Otherwise, `st.markdown()` is used to display the predicted tumor grade in a styled format at the center of the screen.

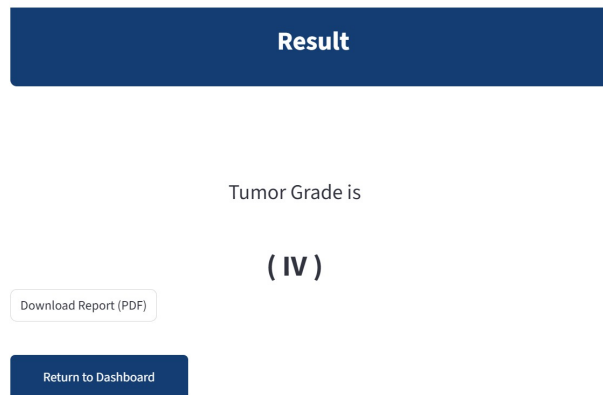


Figure 7.13: Screen of result

```
elif st.session_state.page == "result":
    grade = st.session_state.get("last_grade")
    confidence = st.session_state.get("last_confidence")
    scan_id = st.session_state.get("last_scan_id")

    st.markdown('<div class="center-bar">Result</div>', unsafe_allow_html=True)

    if not grade or not scan_id:
        st.warning("No result available.")
    else:
        st.markdown(f"""
            <div style='text-align:center; font-size:26px; margin-top:40px;'>
                Tumor Grade is <br><br>
                <span style='font-size:36px; font-weight:bold;'>{ (grade) }</span>
            </div>
            """, unsafe_allow_html=True)
```

Figure 7.14: Code of Result Screen

Usability testing was conducted by allowing a group of users to interact with the system. Feedback was collected regarding ease of navigation, clarity of information, and overall user experience. Based on this feedback, improvements were made, including refining button placement, enhancing layout structure, and improving readability. Additional enhancements were also applied to improve responsiveness across different devices and screen sizes, as well as optimizing the arrangement of interface elements for better user interaction.

7.5 Backend Development

The backend is responsible for handling system logic, including database operations, MRI preprocessing, and communication with the deep learning model. It ensures efficient processing and reliable system performance.

Figure 7.11 This code defines the Patient table in the database, which stores essential patient information such as patient ID, record number (MRN), gender, age, and clinical attributes. The record number is set as a unique field to ensure that each patient is uniquely identified. Additionally, the table includes a foreign key that links each patient to a supervising user, maintaining data integrity and relationships within the system.

```
CREATE TABLE IF NOT EXISTS Patient (  
    PatientID INTEGER PRIMARY KEY AUTOINCREMENT,  
    RecordNumber TEXT UNIQUE NOT NULL,      -- Patient ID / MRN  
    Sex TEXT,  
    AgeAtMRI REAL,  
    WHOCNSGrade TEXT,  
    FinalDiagnosisWHO2021 TEXT,  
    MGMTStatus TEXT,  
    IDH TEXT,  
    SupervisingUserID INTEGER,  
    FOREIGN KEY (SupervisingUserID) REFERENCES User(UserID)  
        ON UPDATE CASCADE  
        ON DELETE SET NULL  
);  
""")
```

Figure 7.15: Patient Table

Figure 7.12 This function is responsible for storing MRI scan data in the database. It first establishes a connection to the database and creates a cursor to execute SQL commands. Then, it inserts a new record into the Scan table, including the patient ID, upload date (generated automatically), file path, and original file name. After executing the query, the changes are committed to ensure the data is saved permanently. The function then retrieves the automatically generated scan ID and closes the database connection. Finally, it returns the scan ID so it can be used later to link the scan with analysis results or other operations.

```
def create_scan(patient_id: int, file_path: str, original_file_name: str | None = None) -> int:
    conn = get_conn()
    cur = conn.cursor()
    cur.execute("""
        INSERT INTO Scan (PatientID, UploadDate, FilePath, OriginalFileName)
        VALUES (?, ?, ?, ?)
        """, (patient_id, datetime.now().isoformat(), file_path, original_file_name))
    conn.commit()
    scan_id = cur.lastrowid
    conn.close()
    return scan_id
```

Figure 7.16: Store MRI Scan

Figure 7.13 This function is responsible for storing the prediction results generated by the deep learning model in the database. It begins by establishing a connection and creating a cursor to execute SQL commands. The function then inserts (or replaces) a record in the DiagnosisResult table, including the scan ID, predicted tumor grade, and the timestamp of when the result was created. The use of "INSERT OR REPLACE" ensures that existing results for the same scan can be updated if needed. After saving the data, the transaction is committed to make the changes permanent. The function then retrieves the result ID associated with the scan and returns it, allowing the system to reference the stored prediction in subsequent operations.

```
def save_result(scan_id: int, tumor_grade: str) -> int:
    conn = get_conn()
    cur = conn.cursor()
    cur.execute("""
        INSERT OR REPLACE INTO DiagnosisResult (ScanID, TumorGrade, CreatedAt)
        VALUES (?, ?, ?)
        """, (scan_id, tumor_grade, datetime.now().isoformat()))
    conn.commit()
    cur.execute("SELECT ResultID FROM DiagnosisResult WHERE ScanID = ?", (scan_id,))
    rid = cur.fetchone()["ResultID"]
    conn.close()
    return rid
```

Figure 7.17: Save Prediction Result

Figure 7.14 This code generates the tumor grade prediction using the trained deep learning model. First, the model processes the input MRI image and produces a set of prediction probabilities for each class. Then, the index of the highest probability is selected using the argmax function. Finally, this index is mapped to the corresponding tumor grade label, resulting in the final predicted class.

```

preds = model.predict(img, verbose=0)

grade_index = np.argmax(preds)

grade = labels[grade_index]

```

Figure 7.18: Prediction Code

Figure 7.15 This code performs preprocessing on MRI images before feeding them into the deep learning model. First, the image is resized to a fixed dimension (128×128) to match the model input requirements. Then, intensity normalization is applied using percentile scaling (from 1% to 99%) to enhance contrast and reduce the effect of outliers. Finally, the pixel values are clipped to a range between 0 and 1, ensuring consistent input for the model and improving prediction performance.

```

slice_img = cv2.resize(slice_img, (128, 128))
p1, p99 = np.percentile(slice_img, 1), np.percentile(slice_img, 99)
slice_img = (slice_img - p1) / (p99 - p1 + 1e-8)
slice_img = np.clip(slice_img, 0, 1)

```

Figure 7.19: Image Preprocessing

The backend plays a crucial role in managing system operations, ensuring smooth communication between components, and generating accurate predictions.

7.6 Deployment

Figure 7.16 illustrates the deployment architecture of the proposed system, showing how the system can be hosted on a cloud platform such as AWS, and how its main components, including the user interface, backend services, database, and monitoring tools, interact during system operation. For public deployment, the system can be hosted on a cloud platform such as AWS.

Prepare for Deployment: The system is prepared by structuring it into a frontend interface and a backend service. The backend handles user requests and executes the machine learning model, while the frontend provides an interactive user interface. All required dependencies are defined, and the system is tested locally

to ensure proper functionality. The system is containerized using Docker to ensure consistency across different environments and simplify deployment.

Deploy the System: The application is deployed on a cloud-based environment such as AWS, where a virtual server is configured and the runtime environment is prepared. The system is made accessible through a public web interface. A local SQLite database is used during development, and it can be replaced with a cloud-based database during deployment to support scalability and multiple users. Image data is stored using cloud storage solutions. The deployment process ensures proper configuration of all system components and reliable communication between them.

Monitor Post-Deployment: After deployment, the system is continuously monitored using tools such as AWS CloudWatch to ensure performance, stability, and reliability. Monitoring includes tracking system activity, validating outputs, and detecting errors. Logs are maintained to identify issues related to data processing and system operations. Alerts are configured to notify administrators of any failures, ensuring timely maintenance, minimal downtime, and consistent system performance.

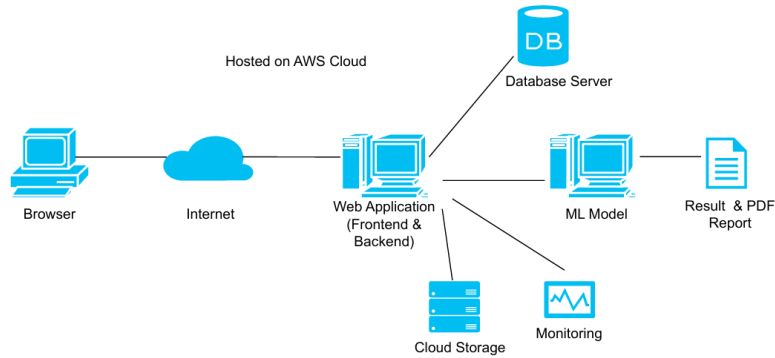


Figure 7.20: System Deployment Diagram

7.7 Summary

This chapter details the implementation process of the proposed brain tumor classification system. It begins with the development of a clear implementation plan, followed by the development and integration of the system's core modules to ensure effective functional integration. The user interface and backend of the system were also developed to guarantee ease of use, accurate data processing,

and efficient operation of deep learning models. Testing and debugging were then conducted to verify the system's integrity and stability, demonstrating its readiness for practical application in medical settings. The deployment phase has shown the system's potential for effectively supporting clinicians in the diagnosis and staging of tumors. Despite the successful implementation, a thorough evaluation of the system's performance is essential to ensure its efficiency and reliability. Therefore, the next chapter moves to the testing and evaluation phase, where the system's performance will be analyzed using a range of different measurement criteria to determine its accuracy and effectiveness in handling MRI data in real-world environments.

Chapter 8

Testing and Evaluation

8.1 Testing Strategy

8.2 Unit Testing

8.3 Integration Testing

8.4 System Testing

8.5 Evaluation of System Performance

8.6 Summary

8 Testing and Evaluation

This project aims to develop a system based on deep learning techniques for staging brain tumors using magnetic resonance imaging (MRI) images. While most current studies only determine the presence or absence of a tumor, this research seeks to provide a precise staging classification, supporting clinicians in designing personalized treatment plans and effectively monitoring disease progression. To ensure the system's reliability and readiness for the clinical environment, the model and software platform underwent a series of comprehensive tests (detailed in Chapter 8), including module testing, data integrity testing, and user acceptance testing, to guarantee the accuracy of the results and minimize human error. The system relies on open-source databases to enhance transparency and the ability to reproduce and further develop the results.

8.1 Testing Strategy

The testing strategy for the proposed brain tumor staging classification system was designed to ensure the reliability, accuracy, and robustness of the system. Several types of tests were used, including Unit Testing, Integration Testing, System Testing, and also User Acceptance Testing (UAT). These stages were carried out sequentially to ensure that each component functions correctly, whether independently or as part of the complete system.

The scope of testing included all the core components of the system, including user authentication (login), database operations, MRI image uploading and processing, as well as tumor classification using a deep learning model, in addition to result generation. Special attention was given to sensitive functions such as login verification, patient number verification, and prediction accuracy, due to their direct impact on system reliability and ease of use.

Unit testing was conducted on individual components such as the login verification function, the image processing function, and the prediction module to ensure that each unit produces the correct results. Integration testing was also carried out to verify the interaction of the user interface with the backend system logic, the database, and the deep learning model, ensuring smooth data flow within the system. System testing included evaluating the complete workflow, starting from logging in, entering the patient number, uploading MRI images, generating predictions, and downloading reports. In addition, UAT was conducted by providing the system to users to try and give their feedback on ease of use and performance.

The testing environment included a local development environment. Initial tests were conducted locally using a development environment to verify functionality and detect errors. Tests were also carried out using a local Streamlit environment through a web browser to simulate real usage conditions. This helped ensure

that the system operates correctly in actual usage scenarios, including handling user input, interacting with the database, and performing real-time predictions.

8.2 Unit Testing

Unit testing was conducted to verify the functionality of individual components of the system, ensuring that each unit performs as expected under various conditions. Each module was tested independently to ensure correctness before integration with other system components. The unit testing process covered core modules of the system, including user authentication (login), database operations, user interface handling (Streamlit application), MRI image preprocessing, prediction functions, and heatmap visualization. Each unit was tested in isolation by providing specific inputs and validating the outputs against expected results.

Examples of test cases:

- Test Case 1: Login Function Input: Valid username and password Expected Output: Successful login and access to the system Actual Output: User logged in successfully.
- Test Case 2: Image Upload Handling Input: MRI image uploaded through the Streamlit interface Expected Output: Image successfully received and passed to preprocessing module Actual Output: Image uploaded and processed correctly.
- Test Case 3: MRI Image Preprocessing Input: Raw MRI image Expected Output: Image resized and normalized to match model input requirements Actual Output: Image correctly resized and normalized.
- Test Case 4: Tumor Stage Prediction Function Input: Preprocessed MRI image Expected Output: Predicted tumor stage (e.g., Stage II, Stage III, or Stage IV) with confidence score Actual Output: Correct tumor stage predicted with confidence score.
- Test Case 5: Heatmap Generation Input: MRI image and prediction result Expected Output: Heatmap highlighting tumor-relevant regions Actual Output: Heatmap generated correctly and aligned with prediction.
- Test Case 6: Database Operation Input: Patient data entered into the system Expected Output: Data stored successfully in the database Actual Output: Data saved and retrieved correctly Unit testing was conducted by executing individual modules such as `predict.py`, `heatmap.py`, and preprocessing functions separately, and verifying their outputs using sample MRI images. Additionally, the Streamlit application (`app.py`) was used to manually test the interaction between components.

The testing process was conducted manually by executing individual modules such as `predict.py`, `heatmap.py`, and database functions separately. Each function was tested using sample inputs, such as MRI images and user data, and the outputs were validated against expected results. In addition, the full system was tested through the Streamlit application (`app.py`), where different scenarios were performed, including user login, image upload, prediction generation, and heatmap visualization. Multiple test cases were applied to ensure system correctness, stability, and reliability.

8.3 Integration Testing

Integration testing was conducted after completing unit testing to ensure that all system components of the brain tumor classification system work together correctly and efficiently. The main objective was to verify the interaction between key modules, including MRI image preprocessing, tumor detection and segmentation, deep learning classification model, and result visualization.

One of the primary integration test scenarios focused on validating the complete workflow from image input to final classification output:

Test Scenario: Integration of MRI Image Processing and Classification Modules

Steps:

1. The user uploads an MRI image into the system.
2. The preprocessing module standardizes the image (normalization, resizing)
3. The segmentation module identifies the tumor region.
4. The processed image is passed to the classification model.
5. The system generates the tumor stage prediction.
6. The result is displayed to the user.

Expected Outcome: The MRI image is successfully processed through all modules, and the system outputs an accurate tumor classification result without errors or delays.

Another important scenario tested the integration between the classification module and the database system:

Test Scenario: Integration of Classification Results and Data Storage

Steps:

1. The system generates classification results after analysis.

2. The results (tumor stage, image data, and metadata) are stored in the SQLite database.

Expected Outcome: All results are correctly stored and retrieved without data loss or inconsistency.

8.4 System Testing

A comprehensive system test was conducted to verify the efficiency of the programmed functions, technical performance, and data security. This included functional tests to ensure the system performed its classification tasks and non-functional tests to evaluate speed and ease of use. The results were documented, and any gaps identified during testing were addressed. System tests were conducted to verify the overall operational efficiency of the brain tumor staging application. This involved testing all use cases from start to finish, from uploading an MRI image to obtaining an accurate stage diagnosis, ensuring the system met all predefined requirements. These tests were conducted based on the functional and non-functional requirements defined in earlier chapters.

The functional test focused on verifying the accuracy of the input image processing and the model's ability to classify tumor stages such as Grade II, Grade III, and Grade IV accurately. The non-functional testing included evaluating response speed when processing high-resolution images and testing the user interface to ensure ease of use by clinicians or technicians. For example, the system underwent intensive accuracy testing using a validation set. The results demonstrated high stability in the accuracy curves, with the system successfully processing images and delivering classification at a very fast response time, meeting the required performance standards in time-sensitive medical environments. The system also underwent reliability testing to ensure the absence of software errors when uploading files in unsupported formats, and the image processing algorithms were optimized to ensure system stability even with varying input image quality.

Some minor issues were identified during testing, such as handling unsupported file formats and variations in MRI image quality. These issues were addressed by implementing input validation and optimizing preprocessing functions to improve system stability and reliability. Basic security testing was also conducted to ensure safe handling of user inputs and prevent common issues such as invalid data entry. Input validation mechanisms were applied to reduce potential risks and ensure secure interaction between the user interface and the database.[30]

8.5 Evaluation of System Performance

The overall performance of the system was evaluated based on the outputs of the trained deep learning models and the accompanying evaluation metrics[15][18]. The results were compared with reference standards in medical image classification, and areas for future improvement were identified. The brain tumor classification system was evaluated based on the key performance metrics identified during the planning phase, including accuracy, precision, recall, and overall user experience. In addition, multiple deep learning models such as CNN, MobileNetV2, and ResNet50 were evaluated[28][18], and the best-performing model was selected based on overall performance.

The results obtained from the evaluation showed that the system achieved high performance, with the CNN model providing the best results among the tested models. The system achieved high accuracy based on the validation results, while the confusion matrix demonstrated the model’s ability to correctly classify tumor stages (Grade II, Grade III, and Grade IV), as shown in Table 8.1. These results indicate that the system can effectively support medical diagnosis and reduce potential human error, and the performance of other models used during training, such as MobileNetV2 and ResNet50, is also presented for comparison.

Table 8.1: Performance Comparison of Deep Learning Models

Models	accuracy	precision	recall
S. Lysdahlgaard et al. 2024 [47]	0.95	-	-
CNN	0.80	0.80	1.00
ResNet50	0.57	0.92	0.59
Mobilenetv2	0.79	0.82	0.97

In contrast to the research by Artificial Intelligence and Advanced MRI Techniques: A Comprehensive Analysis of Diffuse Gliomas [47], which achieved higher accuracy using UCSF-PDGM data, our system was developed with a different objective and a different working mechanism. The project went beyond focusing solely or using feature extraction techniques; instead, we built an integrated end-to-end pipeline that simulates real-world use within a medical setting.

The system allows users to directly upload MRI images, which are then analyzed

using deep learning models. A comprehensive PDF report is then generated, containing the predicted tumor grade, confidence score, and visual interpretations of the model’s decision. We also employed Grad-CAM and Heatmap techniques to highlight the regions where the model focused on the tumor, helping clinicians and radiologists understand the decision-making process and increasing the system’s reliability.

Unlike many studies that rely heavily on feature engineering and manual feature extraction, we deliberately avoided these methods due to their complexity, lengthy processing time, and limited effectiveness in hospital and healthcare settings, which require fast and scalable systems. Therefore, we focused on building a practical and clinically applicable system with a focus on scalability and providing clearly interpretable results for clinicians and radiologists.

Although the accuracy we achieved was lower than the reference research, our system prioritizes practicality, clinical usability, results interpretation, and delivering a comprehensive, usable solution, rather than only achieving high accuracy values.

User feedback indicated that the system is easy to use and provides clear and fast results, which enhances user satisfaction and usability. Compared with previous studies, the use of deep learning architectures proved effective in extracting important tumor features that may not be easily detected manually. The system also showed stable performance during testing. However, some areas for future improvement were identified, such as expanding the dataset to include rare tumor cases and enhancing system integration with medical imaging devices for real-time diagnosis.

8.6 Summary

In conclusion, this research has developed an integrated system capable of accurately classifying brain tumor stages using deep learning models. The test and evaluation results presented in Chapter 8 demonstrate the system’s success in achieving an effective balance between image processing and medical report extraction. The core software functions successfully passed module and integration tests, and the results showed a high degree of predictive power for different tumor stages. Despite the technical challenges encountered during the project, the findings open promising avenues for improving the accuracy of medical diagnosis. We hope this work represents a step towards deeper integration of artificial intelligence into the healthcare system. Future research aims to expand the dataset and enhance the system’s ability to handle more complex medical cases to ensure optimal patient safety outcomes.

Chapter 9

Conclusion

9.1 Summary of Project Objectives

9.2 Key Findings and Results

9.3 Challenges and Solutions

9.4 Contribution to the Field

9.5 Recommendations for Future Work

9 Conclusion

This concluding chapter represents the culmination of the research and development journey for the brain tumor staging system project using deep learning and magnetic resonance imaging (MRI). It aims to provide a comprehensive overview of the project's achievements, from verifying the attainment of the objectives outlined in previous chapters to reviewing the system's key technical and medical findings. The chapter also highlights the challenges faced by the team and how they were overcome through innovative approaches, while demonstrating the added value this research offers in integrating artificial intelligence into the healthcare system. Finally, the chapter presents a set of recommendations and suggestions for future development of this work to ensure enhanced diagnostic accuracy and improved patient care outcomes.

9.1 Summary of Project Objectives

Based on the objectives outlined in Chapter 1, this project aimed to design and implement a deep learning-based system capable of accurately classifying the stages of glioma brain tumors using MRI images. In addition, it sought to provide a practical AI-based tool to assist healthcare professionals in treatment decision-making, monitoring tumor progression, and improving patient outcomes. The project also aimed to demonstrate the effectiveness of deep learning in medical imaging using open-source datasets and to establish a foundation for future research in automated tumor analysis and AI-assisted diagnostics. Throughout the project, these objectives were successfully achieved through key milestones, including data collection and preprocessing, model development and training, and performance evaluation. The system demonstrated reliable classification performance with satisfactory accuracy, confirming its potential as a supportive diagnostic tool and reinforcing the role of artificial intelligence in advancing modern healthcare.

9.2 Key Findings and Results

The project revealed several important results related to the performance and ease of use of the proposed brain tumor staging classification system. The results showed that integrating deep learning techniques with an interactive web interface can effectively support medical image analysis and assist in predicting tumor stages.

Initial tests conducted by team members also indicated a high level of satisfaction with the user interface and the system's functions in general, especially the ease

of the login process, the simplicity of uploading MRI images, and the clarity of the displayed results. It was also found that the system is easy to use and navigate during the tests.

System tests showed that the application is capable of efficiently processing MRI images and generating tumor classification results within a short response time. The selected deep learning model (CNN) achieved the best performance compared to the other models tested, demonstrating high accuracy in classifying tumor stages (grade two, three, and four). The system also maintained consistent performance during testing, with reliable data flow between the user interface, the backend, the database, and the prediction model.

One of the unexpected challenges faced by the project was handling large 3D MRI images and preparing them for input into the model. This issue was resolved by extracting the middle slice of the MRI image and applying preprocessing techniques such as resizing and normalization to meet the model's requirements. Another challenge encountered was ensuring seamless integration between system components, which was addressed through repeated testing and bug fixing.

In general, these results confirm that the system is capable of providing accurate, reliable, and user-friendly classification of brain tumors within the test environment, with reference to future opportunities for improvement such as expanding the dataset and increasing the system's scalability.

9.3 Challenges and Solutions

One of the main challenges faced by the project was dealing with large 3D MRI images and preparing them to be suitable for input into the deep learning model. MRI images are usually stored as volumetric data, making it difficult to process them directly using models based on 2D images. To overcome this, a preprocessing method was applied, where the central slice of the MRI image was extracted and resized to match the input dimensions required by the model. In addition, normalization techniques were applied to standardize the pixel intensity values, ensuring their compatibility with the trained model.

Another challenge was represented in integrating the different components of the system, including the user interface built using Streamlit, the backend logic, the database operations, and the deep learning model. Ensuring smooth communication and data flow between these components required precise debugging and repeated testing with error optimization during development. This challenge was addressed by dividing the code into independent functions and files (such as prediction modules, preprocessing, and database handling), in addition to verifying each component separately before integrating it into the full system.

An additional challenge was choosing the most suitable deep learning model for classifying tumor stages. Several models were trained and evaluated, including

CNN, MobileNetV2, and ResNet50, using performance metrics such as Accuracy, Precision, Recall, and F1 score. Based on the evaluation results, the CNN model was selected as the final model due to its superior performance.

One of the challenges encountered was the weak visualization performance of the CNN model when applying Grad-CAM. To solve this issue, ResNet50 was integrated specifically for heatmap generation, significantly improving the interpretability of the model outputs.

These challenges have provided valuable learning experiences, highlighting the importance of properly preprocessing data, designing the system in a modular way, and evaluating models systematically. They also emphasized the importance of conducting repeated tests and continuous verification to ensure the system's reliability and stability.

9.4 Contribution to the Field

This project makes a significant contribution to the field of computer science, particularly to artificial intelligence applications in medicine, by developing a system based on deep learning techniques for staging brain tumors using (MRI) images. Unlike many traditional systems that focus solely on detecting the presence of a tumor, this project offers a more advanced solution by emphasizing the precise classification of tumor stages, thus supporting clinicians in making more accurate and effective treatment decisions.

A key contribution of this work is a practical comparison of several deep learning models, such as CNN, ResNet, and MobileNetV2, to evaluate their performance in terms of accuracy and efficiency. The results of the experiments showed that the CNN model achieved the best performance in terms of accuracy and stability, making it the most suitable choice for adoption in the proposed system.

The project also features the integration of explainable AI techniques, such as Grad-CAM, which enhances the model's transparency and helps clinicians understand the decision-making process, thereby increasing the system's reliability and dependability in medical settings. Furthermore, techniques such as data augmentation were employed to improve the model's generalization ability and mitigate the problem of data diversity.

This project contributes to providing a practical solution applicable in real-world clinical settings, combining high accuracy with operational efficiency. This supports the development of intelligent diagnostic systems that help reduce human error and improve the quality of healthcare. In the long term, this work forms a foundation upon which future research in medical image analysis using artificial intelligence can be built.

9.5 Recommendations for Future Work

Although the current system achieves the project’s core objectives, several areas offer potential for future development. Key improvements include integrating more advanced deep learning models or employing hybrid models to enhance classification accuracy and system performance.

Expanding the dataset by incorporating real-world clinical data from multiple sources is also recommended. This will improve the model’s generalizability across real-world settings. Furthermore, developing a web or mobile application will increase accessibility and usability for healthcare professionals in diverse environments.

Another important future direction is the integration of multimodal data, such as combining MRI images with clinical and genetic data, to improve prediction accuracy. Real-time processing and cloud deployment can also be implemented to make the system more scalable for large healthcare institutions.

Finally, future research can focus on developing more interpretable AI technologies to provide clearer and more in-depth interpretations, thereby increasing physicians’ confidence in the system and supporting more accurate and reliable medical decisions.

Chapter 10

References

An example of the generated system report is shown in Figure 9.1, which illustrates the patient information, predicted tumor grade, confidence score, and the corresponding heatmap visualization produced by the model.

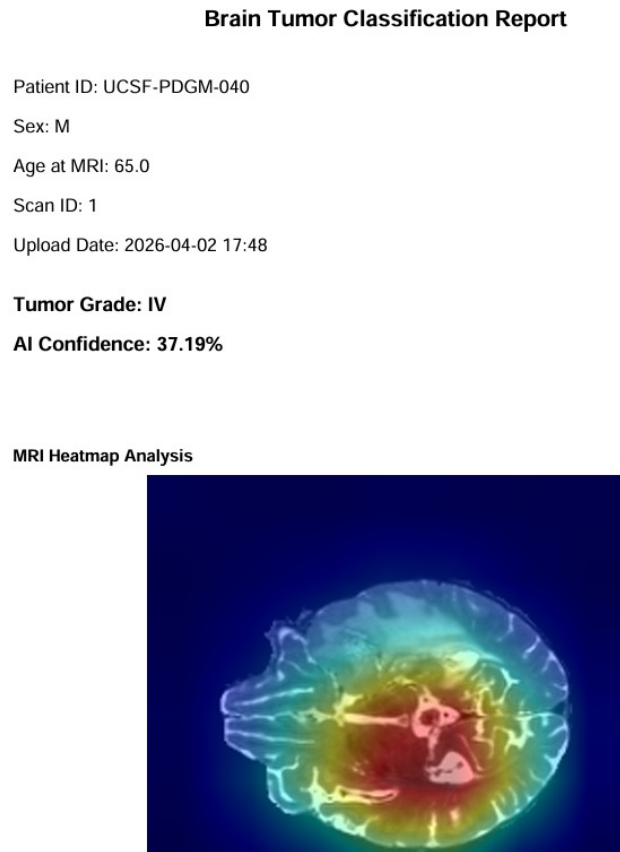


Figure 9.1: System-Generated Brain Tumor Classification Report

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